



DETAILED PROJECT REPORT (DPR)

SUPPLY CHAIN MANAGEMENT

Contents

1. PREAMBLE	4
2. AGRICULTURE RESIDUES AVAILABILITY OVERVIEW	6
3. AGRICULTURE RESIDUES PROCESSING TECHNOLOGIES	18
4. BIOCHAR SEQUESTRATION OVERVIEW	21
5. PROJECT OBJECTIVES & SCALABLE MODEL	25
6. SKILLING & GREEN ENTREPRENEURS DEVELOPMENT	29
7. GREEN ENTREPRENEURS MENTORING & OPERATIONS MANAGEMENT	33
8. PROJECT COMMAND AREA	35
9. FEEDSTOCK SUPPLY CHAIN MANAGEMENT - AT SOURCE	44
10. TRANSPORT & LOGISTICS	52
11. FEEDSTOCK MATERIAL RECEIPT & PROCESSING	57
12. PROCESSED FEEDSTOCK STORAGE	61
13. PYROLYSER OUTPUT - SUPPLY CHAIN MANAGEMENT	66
14. DIGITAL M.R.V. & DASHBOARD	68
15. BIO-OIL	73
16. POWER SOURCING & ELECTRICAL BALANCE OF PLANT (BOP)	76
17. AUTOMATION & INSTRUMENTATION	79
18. CIVIL & STRUCTURAL WORKS; BUILDINGS	82
19. HEALTH, SAFETY & ENVIRONMENT MANAGEMENT	85
20. ORGANIZATION – PLANT CONSTRUCTION AND O&M	98
21. IMPLEMENTATION SCHEDULE	101
22. PROJECT COST BREAKDOWN	105
23. PROJECT FINANCIALS OVERVIEW REQUIRED	108
24. RISK ASSESSMENT & MITIGATION PLAN	111
25. ENVIRONMENTAL, ECOLOGICAL & SOCIAL IMPACT	116

PREAMBLE

1. PREAMBLE

Project Context & Rationale

The agricultural heartland of Karnataka, specifically the corridor encompassing **K R Pet, Nagamangala, Pandavapura, Channarayapatna, and Arsikere**, generates over **500,000 tons** of agricultural residue annually. Currently, a significant portion of this biomass—specifically **coconut fronds, sugarcane trash, and paddy stalks**—is managed through open-field burning or inefficient decomposition.

This project seeks to transition this "linear waste liability" into a "circular economic asset" by establishing a robust, technology-driven supply chain to feed a Phase 1: 2×1 TPH modular biochar plant (scaling to 12 TPH by FY 2028-29).

The Problem Statement

Despite the abundance of biomass, large-scale biochar production often fails due to the **"Logistics Paradox"**:

- **Low Bulk Density:** Raw agricultural waste is mostly "air," making transport over long distances financially unviable.
- **Seasonality:** Harvesting windows for paddy and cane are narrow (15–30 days), leading to supply shocks.
- **Traceability Gap:** Without a digital trail, the resulting biochar cannot qualify for high-value International Carbon Removal Credits (CORCs).

Strategic Objectives

The supply chain framework outlined in this document is designed to achieve three core pillars of operational excellence:

1. **Volume Certainty:** Securing a consistent inflow of **140 to 300 TPD** (Tons Per Day) of feedstock.
2. **Cost Compression:** Utilizing at-source shredding and baling to maximize truck payloads within a **25 km "Golden Radius."**
3. **Quality Standardization:** Delivering feedstock with a consistent moisture profile (<20%) and uniform particle size (<10 mm) to ensure optimal pyrolyser performance.

Scope of the Document

This DPR serves as a technical and financial blueprint for the **Phase 1 (2 TPH)** launch and the **Phase 3 (12 TPH)** scale-up. It details the formation of village clusters, the deployment of mobile processing fleets, and the implementation of a **Logistics Control Tower** to manage the flow of biomass from the farmer's field to the plant's storage bunkers.

Alignment with Global Standards

This project is designed to be compliant with **Verra, Puro.earth** and **European Biochar Certificate (EBC)** standards. By formalizing the collection of waste from smallholder farmers in the Mandya-Hassan region, the project creates a verifiable "Carbon Sink" that contributes to India's **Net Zero 2070** commitments while providing a 15–20% income boost to local farming communities.

AGRICULTURE RESIDUES AVAILABILITY OVERVIEW

2. AGRICULTURE RESIDUES AVAILABILITY OVERVIEW

A. Estimated annual sustainable feedstock composition:

Feedstock Type	% Share	Characteristics
Coconut residues	35–40%	High lignin content; slow pyrolysis friendly
Sugarcane trash & bagasse	35–40%	Readily available; high annual turnover
Paddy husk, maize stalks, horticulture waste	15–20%	Supplementary feedstocks
Bamboo residues	10–20%	Enhances carbon stability

Sustainable feedstock characteristics:

- All feedstocks are residual in nature, not purpose-grown
- Locally available
- Non-competitive with food or fodder uses
- No deforestation or land-use change involved

This diversified residue basket reduces supply risk, stabilises carbon yield, and optimises the plant's operational efficiency.



B. Seasonal Availability & Collection Risk Calendar

The chart below shows the availability of biomass throughout the year.

- **Green:** Peak availability (Harvest season - lowest price).
- **Yellow:** Moderate availability (Post-harvest clearing).
- **Red:** Low/No availability (Planting/Growing season - storage required).

Biomass Type	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Coconut Fronds	●	●	●	●	●	●	●	●	●	●	●	●
Sugarcane Trash	●	●	●	●	●	●	●	●	●	●	●	●
Paddy Straw	●	●	●	●	●	●	●	●	●	●	●	●
Bamboo Residue	●	●	●	●	●	●	●	●	●	●	●	●

Biomass Availability by Taluka*

Taluka	Biomass Type	Area (Acres)	Yield (MT/Acre)	Total Availability (MT)	Usage by Others (40%)	Surplus Quantity (MT)
K R Pet	Cane Trash	29,500	4.5	1,32,750	53,100	79,650
K R Pet	Coconut Fronds	34,500	3	1,03,500	41,400	62,100
K R Pet	Paddy Trash	26,500	1.8	47,700	19,080	28,620
K R Pet	Maize/Other	22,000	2	44,000	17,600	26,400
Pandavapura	Cane Trash	36,000	4.5	1,62,000	64,800	97,200
Pandavapura	Paddy Trash	38,000	1.8	68,400	27,360	41,040
Pandavapura	Coconut Fronds	13,000	3	39,000	15,600	23,400
Pandavapura	Other Agri-Waste	8,500	1.2	10,200	4,080	6,120
Nagamangala	Cane Trash	11,500	4.5	51,750	20,700	31,050
Nagamangala	Coconut Fronds	26,000	3	78,000	31,200	46,800
Nagamangala	Ragi/Other	28,000	1.5	42,000	16,800	25,200
Nagamangala	Paddy Trash	14,000	1.8	25,200	10,080	15,120
Channarayapatna	Coconut Fronds	58,000	3	1,74,000	69,600	1,04,400
Channarayapatna	Maize/Other	48,000	2.5	1,20,000	48,000	72,000
Channarayapatna	Cane Trash	14,000	4.5	63,000	25,200	37,800
Channarayapatna	Paddy Trash	12,500	1.8	22,500	9,000	13,500

*Data Source Summary

- **Horticulture Data:** District-wise area & production stats (2024-25 Forecast), Govt. of Karnataka.
- **Cane Data:** Directorate of Sugar and Cane Development, Karnataka (Mill-wise catchment areas).
- **Yield Ratios:** Based on ICAR-CPCRI (Coconut), ICAR-SBI (Sugarcane), and UAS-Bangalore (Cereals) residue-to-grain/fruit ratios.

C. Coconut fronds collection & transportation

Efficiently converting coconut fronds into biochar feedstock requires a structured process to manage their bulky, fibrous nature. Since a single frond can be up to 6 meters long, processing at or near the farm is critical to making transportation economically viable.

1. Collection & Field Management

- **Natural Drop & Pruning:** An adult palm produces approximately 12 to 14 new leaves annually, at a fairly regular interval of about one leaf and an inflorescence (flower cluster) every month. A single coconut leaf stays on the palm for about two to two and a half years before it naturally senesces (dries out) and falls. The number of leaves shed annually roughly equals the number of new leaves produced, maintaining a consistent crown size of around 30-40 leaves at any given time. Fallen coconut fronds should be cleared twice a year, to proactively manage hazards and pests
- **Aggregation:** trained local green entrepreneurs aggregate the fronds at a central point in the farm. Heavy-duty garden rakes, with metal tines or adjustable heads, will be used to effectively gather tough fronds into manageable piles for collection
- **On-Site Drying:** Fronds are left to air-dry for **10–15 days**. This reduces moisture from ~50% to <20%, which is critical because transporting "water weight" is the biggest hidden cost in biomass logistics.

2. Shredding & Volume Reduction

- **Mobile Shredding:** tractor-PTO-operated or diesel-powered mobile shredders are brought to the plantation for shredding / pulverising at source
- **The Process:** The long frond (including the thick woody petiole/base) is fed into the shredder, which uses high-speed blades (typically 15–35 HP machines) to break it down into **20mm to 50mm chips**.
- **Density Gains:** Shredding increases the bulk density from roughly 50 kg/m³ (loose fronds) to over 250 kg/m³ (chips), allowing a single truck to carry **5x more material**.

Coconut Frond Collection



Shredding



Powdered coconut frond collection on tarps



3. Transportation to the Biochar Plant

- Tiller trolley capacity of 1,000 Kg will be used for moving the gathered fronds from the field to the processing area for shredding
- **Mechanized Loading:** Shredded coconut fronds are blown directly into a tarp covered high-walled tractor-trailers via a discharge chute, eliminating manual bagging labor. Alternatively, large, reinforced tarp with handles into which fronds are raked into and then use the handles to easily lift and dump the load into cart or trailer
- **Logistics Clustering:** To keep costs low, collection is done in "clusters." A truck visits 3–4 neighboring farms in a single trip to reach its full capacity.

Transportation of powdered coconut frond from source to biomass storage at biochar plant



4. Delivery, Quality Check & Storage

- **Weighment:** The vehicle is weighed at the biochar plant's weighbridge.
- **Moisture Testing:** Upon arrival at the biochar plant, 2 samples will be collected & the moisture is verified. If the coconut fronds were shredded & dried in the field, they can be fed directly into the continuous pyrolysis reactor without further energy-intensive drying
- **Storage:** If the moisture content of the powdered coconut frond is more than the specified range the powdered coconut frond will be stored in a shed & dried to reduce the moisture content.

Weighment



D. Cane trash collection

Managing sugarcane trash is one of the most labor-intensive but rewarding biomass processes due to the sheer volume available in the Mandya "Sugar Belt." In this region, trash is often traditionally burnt, so a structured collection process is vital for carbon sequestration.

1. Field Collection & Windrowing

- **Post-Harvest Drying:** After the cane stalks are cut and sent to the sugar mill, the leaves and tops (trash) are left in the field for **3 to 7 days** to sun-dry.
- **Manual Raking:** Heavy-duty garden rakes, with metal tines or adjustable heads, will be used to effectively pull the scattered leaves into long, organized rows called **windrows**.
- **Separation:** During raking, care is taken to exclude soil and stones, which can damage shredding blades and increase the ash content of the biochar.

2. Baling (Volume Reduction)

- **Baling for Logistics:** Because shredded trash is extremely light and prone to flying away during transport, many operations use a **Square or Round Baler**. Baling compresses the loose trash into dense blocks, increasing the weight per truckload from 2 tons (loose) to **6–8 tons (baled)**.

3. Transportation to the Biochar Plant

- **Bulk Loading:** Bales weigh <20 kg and are stacked onto the tractor trolley manually.
- **Net Payload:** Transporting baled sugarcane trash is the most cost-effective method, as it maximizes the legal weight limit of the transport vehicle.

4. Delivery, Quality Check & Storage

- **Weighment:** The vehicle is weighed at the biochar plant's weighbridge.
- **Moisture Testing:** Upon arrival at the biochar plant, 2 samples will be collected & the moisture is verified. If the coconut fronds were shredded & dried in the field, they can be fed directly into the continuous pyrolysis reactor without further energy-intensive drying
- **Storage:** If the moisture content of the cane trash bale is more than the specified range it will be stored in a shed & dried to reduce the moisture content.

Raking by collection aids (Manual/Semi-Mechanical) & Baling





Transportation of cane trash from source to biomass storage at biochar plant



E. Bamboo collection & transportation

Efficiently moving bamboo from the plantation to a biochar plant requires a systematic "Selection-Harvest-Processing" workflow to ensure sustainability and minimize transport costs.

1. Selection & Cutting Process

- **Selective Harvesting:** Only mature culms (**3 to 5 years old**) are harvested. Younger culms are left to provide structural support to the clump and ensure future growth.
- **Low-Ground Cutting:** Culms are cut as close to the ground as possible (leaving only one internode) using sharp saws, hand held mechanical cutters to prevent splintering and promote healthy rhizome regeneration.
- **Cleaning:** Side branches and leafy tops are removed on-site. These "leftovers" can be mulched back into the soil or bundled separately as lower-grade biomass.
- **Cleaning the Clump:** Dead, dry, or congested bamboo is removed during the harvest to improve airflow and productivity for the next season.

2. Primary Processing (At the Farm Gate)

- **Sizing/Cross-Cutting:** For logistics efficiency, long bamboo poles (often 10m+) are cut to length of **1-meter** or "slat" forms to fit securely onto transport vehicles.
- **Air Drying:** Harvested bamboo is often stacked vertically in a shaded area for **1 to 2 weeks**. This reduces moisture content (from ~50% to ~25%), significantly lowering the transport weight and fuel costs.

3. Transportation to the Biochar Plant

- **Loading:** Bamboo is bundled and loaded onto tractor-trailers or 10-ton trucks. Because bamboo is hollow and "bulky," maximizing density is key.
- **Volume Optimization:** In some advanced operations, a **mobile chipper** is brought to the farm. Shredding the bamboo into chips at the source increases the "bulk density," allowing a single truck to carry significantly more biomass per trip.
- **Logistics Timing:** Transport is typically scheduled during the **dry months (October–May)** to avoid the weight of rainwater and the difficulty of navigating forest/farm tracks during the monsoon.

4. Delivery, Quality Check & Storage

- **Weighment:** The vehicle is weighed at the biochar plant's weighbridge.
- **Moisture Testing:** Upon arrival at the biochar plant, 2 samples will be collected & the moisture is verified. Pyrolysis-grade bamboo ideally requires **<20% moisture** for optimal biochar yield and energy efficiency.
- **Grinding for size reduction:** If the bamboo were slit & dried in the field, they will be further processed in the Chipper & hammer mill / grinder to reduce the size to 2 mm to 5 mm before they can be fed into the continuous pyrolysis reactor without further energy-intensive drying or it will be stored in the shed for drying to achieve the moisture content of <20%.
- **Storage:** If the moisture content of the slit bamboo is more than the specified range the bamboo will be stored in a shed & dried to reduce the moisture content.

Bamboo plant cutting & collection





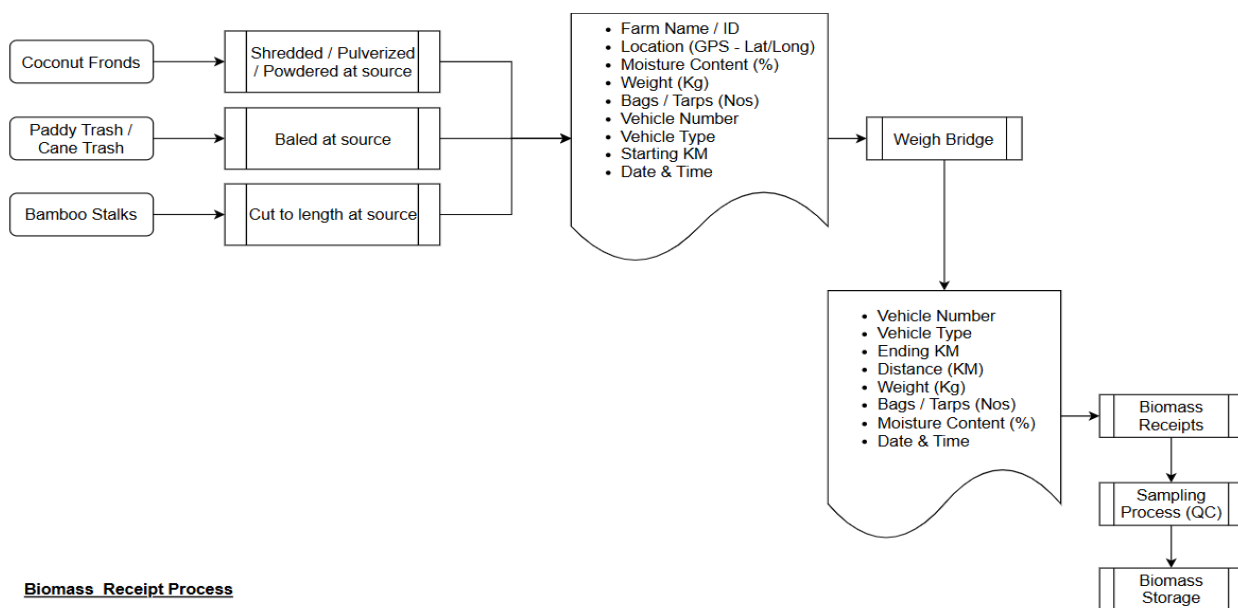
Bamboo cut to length of 1 meter slats



Bamboo stalk transportation to the processing unit & Chipping process



Biomass Receipt



Harvesting frequency: Coconut Fronds – year-round, Paddy Trash – 2 times per year, Cane Trash – 1 time per year

Taluka-wise Residue Availability

Taluka	Residue	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Mandya	Sugarcane Trash/Bagasse	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	Yes	Yes
Mandya	Coconut Fronds	Yes	Yes	Yes	No	No	No	No	No	No	Yes	Yes	Yes
Mandya	Paddy Trash	Yes	No	No	No	No	No	No	No	No	Yes	Yes	Yes
Mandya	Cotton Stalk	Yes	No	No	No	No	No	No	No	No	Yes	Yes	Yes
Mandya	Pulses Stalk	Yes	Yes	Yes	No	No	No	No	Yes	Yes	Yes	Yes	No
Mandya	Bamboo	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Krishnarajpet	Sugarcane Trash/Bagasse	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	Yes	Yes
Krishnarajpet	Coconut Fronds	Yes	Yes	Yes	No	No	No	No	No	No	Yes	Yes	Yes
Krishnarajpet	Paddy Trash	Yes	No	No	No	No	No	No	No	No	Yes	Yes	Yes
Krishnarajpet	Cotton Stalk	Yes	No	No	No	No	No	No	No	No	Yes	Yes	Yes
Krishnarajpet	Pulses Stalk	Yes	Yes	Yes	No	No	No	No	Yes	Yes	Yes	Yes	No
Krishnarajpet	Bamboo	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nagamangala	Sugarcane Trash/Bagasse	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	Yes	Yes
Nagamangala	Coconut Fronds	Yes	Yes	Yes	No	No	No	No	No	No	Yes	Yes	Yes
Nagamangala	Paddy Trash	Yes	No	No	No	No	No	No	No	No	Yes	Yes	Yes
Nagamangala	Cotton Stalk	Yes	No	No	No	No	No	No	No	No	Yes	Yes	Yes
Nagamangala	Pulses Stalk	Yes	Yes	Yes	No	No	No	No	Yes	Yes	Yes	Yes	No
Nagamangala	Bamboo	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pandavapura	Sugarcane Trash/Bagasse	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	Yes	Yes
Pandavapura	Coconut Fronds	Yes	Yes	Yes	No	No	No	No	No	No	Yes	Yes	Yes
Pandavapura	Paddy Trash	Yes	No	No	No	No	No	No	No	No	Yes	Yes	Yes
Pandavapura	Cotton Stalk	Yes	No	No	No	No	No	No	No	No	Yes	Yes	Yes
Pandavapura	Pulses Stalk	Yes	Yes	Yes	No	No	No	No	Yes	Yes	Yes	Yes	No
Pandavapura	Bamboo	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Consolidated Biomass Summary Dashboard (FY 2024-25)*

Taluk	Coconut Fronds (MT)	Paddy Trash (MT)	Cane Trash (MT)	Other Agri-Waste (MT)	Total Annual Surplus (MT)
Channarayapatna	52,200	4,500	37,800	48,000	142,500
K R Pet	31,050	9,540	79,650	11,000	131,240
Pandavapura	11,700	13,680	97,200	2,040	124,620
Nagamangala	23,400	5,040	31,050	4,200	63,690
TOTAL	118,350	32,760	245,700	65,240	462,050

***Data Source Summary:** Consolidated from Karnataka Horticulture Dept (FY24-25 Forecast), KSDA Sowing Reports (Dec 2024), and MNRE Biomass Power Atlas.

Footnote: MRPL Report

https://www.google.com/url?sa=i&source=web&rct=j&url=https://environmentclearance.nic.in/writereaddata/Online/TOR/06_Jul_2017_11_015112711JV3TEJMRPLBiomassAssessment.pdf&ved=2ahUKEwi8iranoamSAXSzDgGHeZYMQUQy_kOegQIARAB&opi=89978449&cd&psi=g=AOvVaw0vcNdoDeICeH2qdu4Qy4&ust=1769518787782000

***AGRICULTURE
RESIDUES
PROCESSING
TECHNOLOGIES
OVERVIEW***

3. AGRICULTURE RESIDUES PROCESSING TECHNOLOGIES

3.1 Overview and Technology Selection Criteria

The primary objective of at-source processing is to reduce the bulk volume of agricultural residues before transportation, thereby maximising the payload per truck trip and making the 15–50 km logistics corridor financially viable.

The three key technical targets are:

- (1) Particle size reduction to 5–20 mm for direct pyrolysis feed, or baling for intermediate transport and subsequent grinding at the plant
- (2) Moisture reduction to below 20% (preferably 12–15%) through natural sun-drying or mechanical drying at source
- (3) Density improvement from 50–80 kg/m³ (loose biomass) to 150–300 kg/m³ (shredded) or 400–600 kg/m³ (baled), enabling a 4–8× increase in effective truck payload.

Technology selection is feedstock-specific because coconut fronds, sugarcane trash, paddy straw, and bamboo each require different primary processing equipment due to their distinct fibre length, moisture content, density, and structural toughness.

3.2 Feedstock-wise Processing Technology

Coconut Fronds: Primary equipment is a tractor-PTO-driven or diesel-engine-driven mobile disc/drum chipper (15–35 HP). Fronds (up to 6 m long) are fed rachis-first into the chipper, which reduces them to 20–50 mm chips. Bulk density improves from ~50 kg/m³ (loose fronds) to ~220 kg/m³ (chips), a 4× payload gain. After chipping, fronds are air-dried for 10–15 days at the farm before transport. At the plant, chips are fed through a hammer mill to achieve the final 5–10 mm size for reactor feed.

Indicative equipment: Ecostan/BRT make tractor-PTO chipper, throughput 0.5–1.5 TPH, power 35–55 HP tractor, cost approximately Rs 2.5–4 lakhs.

Sugarcane Trash: Primary equipment is a compact round baler (35–55 HP tractor-drawn). After the cane stalks are harvested and trash is raked into windrows using a side-delivery rake, the baler compresses loose trash (bulk density ~60 kg/m³) into dense cylindrical bales of 400–600 kg each (bulk density ~400 kg/m³), a 6–7× payload gain. Bales are loaded onto flat-bed tractor trailers. At the plant, bales are broken open and fed through a slow-speed twin-shaft shredder before entering the hammer mill. Indicative equipment: Kubota/John Deere compact round baler, throughput 3–5 bales/hour, power 40–55 HP tractor, cost approximately Rs 8–14 lakhs.

Paddy Straw: similar process to cane trash — windrow rake followed by round baler. However paddy straw is more prone to shattering (loss of 5–7% as dust during transport), so bales must be tightly packed and wrapped in net to minimise losses. At the plant, paddy straw is processed through a hammer mill with a 10 mm screen. Moisture control is critical: paddy straw above 20% moisture silages rapidly; bales must be stacked on raised plinths with tarpaulin covering.

Bamboo: primary equipment is a mobile disc chipper (3–5 HP electric or diesel, suitable for 1-metre slat lengths). Bamboo culms (cut to 1 m lengths at the farm) are fed into the chipper to

produce 30–50 mm chips, then stacked for 2 weeks of field drying before transport. At the plant, bamboo chips are processed through the same hammer mill as coconut fronds.

3.3 At-Source Processing Fleet — Phase 1 Requirement (2 TPH)

To sustain the 2-TPH (48 T/day, ~1,200 T/month) inflow, the Phase 1 at-source fleet requirement is: 3 mobile coconut frond chippers (each ~1 TPH throughput, servicing K R Pet cluster on a rotation basis); 2 round balers with matching side-delivery rakes (servicing Pandavapura/Channarayapatna cane trash and paddy straw); 4 tractor-trailers (each 8–10 T capacity) for short-haul (<20 km) logistics from aggregation points to plant; and 1 Light Commercial Vehicle (LCV, 3–5 T) for long-haul sampling runs and moisture meter teams. IoT moisture meters: 6 handheld NIR meters (Bluetooth-enabled, one per cluster supervisor) for field-side moisture verification before collection approval. Total indicative CapEx for Phase 1 at-source fleet: Rs 1.2–1.8 Crore.

BIOCHAR SEQUESTRATION OVERVIEW

4. BIOCHAR SEQUESTRATION OVERVIEW

4.1 What is Biochar Sequestration?

Biochar sequestration is the process of converting agricultural residues into a stable, highly recalcitrant form of carbon — biochar — through oxygen-limited thermal decomposition (pyrolysis), and permanently incorporating this carbon into agricultural soils. Unlike biological carbon (leaves, roots, compost), which decomposes and re-releases CO₂ within 1–10 years, biochar's aromatic carbon structure persists in soil for hundreds to thousands of years. This makes it one of the few genuinely permanent, verifiable Carbon Dioxide Removal (CDR) pathways available at agricultural scale.

4.2 Biochar Application Protocol

Application Rate: 2 tonnes of biochar per hectare per application cycle.

Target crops: paddy (Kharif and Rabi seasons) and sugarcane (annual ratoon cycle) on fields within K R Pet, Pandavapura, Nagamangala, and Channarayapatna talukas.

Application Method: biochar is broadcast and incorporated into the top 0–15 cm of soil using a tractor-mounted rotavator or hand-operated till, applied at the start of the cultivation season before transplanting. This ensures maximum root-zone contact and avoids surface runoff losses.

Timing: paddy fields — application during field preparation in June (Kharif) and December (Rabi); sugarcane fields — application at time of planting or at ratoon initiation in October–November.

Barter Integration: the application of biochar is linked directly to the barter agreement. Farmers who supply biomass feedstock to the plant receive biochar in exchange at a ratio of 1 tonne of biochar for every 4 tonnes of dry biomass delivered. Additionally, a cash payment component compensates farmers for collection and transport effort above the barter equivalent, incentivising timely and quality delivery.

Supplementary Inputs: biochar is co-applied with wood vinegar (diluted 1:300 with irrigation water) as a root bio-stimulant, enhancing soil microbial colonisation and improving the agronomic benefit per hectare for participating farmers.

4.3 Phase-wise Sequestration Targets and Land Coverage

At an application rate of 2 T/ha, the hectares covered per year equals biochar output (TPA) divided by 2. For Phase 1 (Module 1 only, 4,800 TPA biochar at steady state): land coverage = 2,400 hectares per year; gross CDR = $4,800 \times 2.02 = 9,696$ tCO₂/year; less project emissions 725 tCO₂; net CDR = approximately 8,971 tCO₂/year per module. At full scale (6 modules, 28,800 TPA biochar, FY 2029–30): land coverage = 14,400 hectares/year; gross CDR = 58,176 tCO₂/year; net CDR = approximately 53,826 tCO₂/year across the 6-module cluster. This significantly exceeds the 40,000 tCO₂/year target stated in Section 8, which is based on an earlier lower yield assumption. The project will register Modules 1–3 ($3 \times 4,800$ TPA = 14,400 TPA biochar) under a single Verra VCS project for the first crediting period, targeting approximately 26,895 tCO₂ net removal per year at full 3-module operation. Modules 4–6 will be added as a grouped project expansion in FY 2028–29.

4.4 MRV Traceability Chain for Sequestration

The sequestration MRV chain is farmer-driven and barter-linked, creating a self-reinforcing traceability loop.

Step 1 — Feedstock Origin: each biomass delivery from a farmer is logged by village, weight, moisture, and GPS coordinates at the plant weighbridge. A unique Batch ID is assigned.

Step 2 — Production Linkage: biochar produced from each batch is tracked against the input feedstock. The QR code on each 50 kg biochar bag encodes: Batch ID, production date, pyrolysis temperature, organic carbon content (from lab certificate), and destination field parcel.

Step 3 — Barter Dispatch: when a farmer receives biochar under the barter agreement, the MRV app (operated by GAM-trained technicians) scans the bag QR code and links it to the farmer's registered land parcel (GPS-tagged, area in hectares, crop type).

Step 4 — Application Event: the application event is photographed, GPS-tagged, and date-stamped by the field technician using the MRV mobile app. Quantity applied per parcel is recorded.

Step 5 — Soil Sampling: baseline soil organic carbon (SOC) tests are conducted on each registered parcel before first application (using NABL-accredited laboratory, ASTM D2974 or equivalent). Periodic re-testing is conducted at 6 and 12 months post-application to document SOC improvement. All data is stored on the blockchain-enabled DMRV platform described in Section 18.

Step 6 — Carbon Credit Issuance: the aggregated sequestration data (biochar applied, H:Corg, fC_y, GPS parcels, application photos, SOC test certificates) is compiled into a Verification Report submitted to the appointed Verra-accredited third-party auditor annually. Carbon Removal Credits (CORCs) are issued upon successful verification and deposited into the VCS registry account held by the project proponent.

4.5 Agronomic Benefits of Sequestration in Mandya Soils

Mandya's predominantly red laterite and black cotton soils respond well to biochar amendment. Documented benefits in comparable agro-climatic zones include: Water holding capacity (WHC) improvement of 15–25% in sandy-loam soils — critical for K R Pet's rain-fed paddy during the July–September monsoon gap. Cation Exchange Capacity (CEC) increase of 10–20%, reducing leaching of potassium and nitrogen from KRS canal-fed sugarcane soils and lowering fertilizer input cost by an estimated 8–12% per acre.

Soil pH buffering: biochar's alkaline pH (8.5–9.5 for coconut frond biochar) partially offsets the acidification caused by intensive urea application in Mandya's paddy tracts. Microbial habitat: biochar's high porosity (internal surface area 200–400 m²/g) provides refuge for mycorrhizal fungi and nitrogen-fixing bacteria, improving soil biology scores by an estimated 20–30% within two crop cycles.

Reduced chemical runoff: the Hemavathi and KRS canal network drains into the Cauvery River. Biochar-amended soils in the Pandavapura cane belt have measurably lower nitrate and phosphate runoff, contributing to downstream water quality.

Reference: Ippolito et al. (2020), GCB Bioenergy; IBI Biochar Standards v2.1; ICAR-CPCRI Technical Bulletin on Biochar Application in Tropical Soils (2023).

PROJECT OBJECTIVES & SCALABLE MODEL

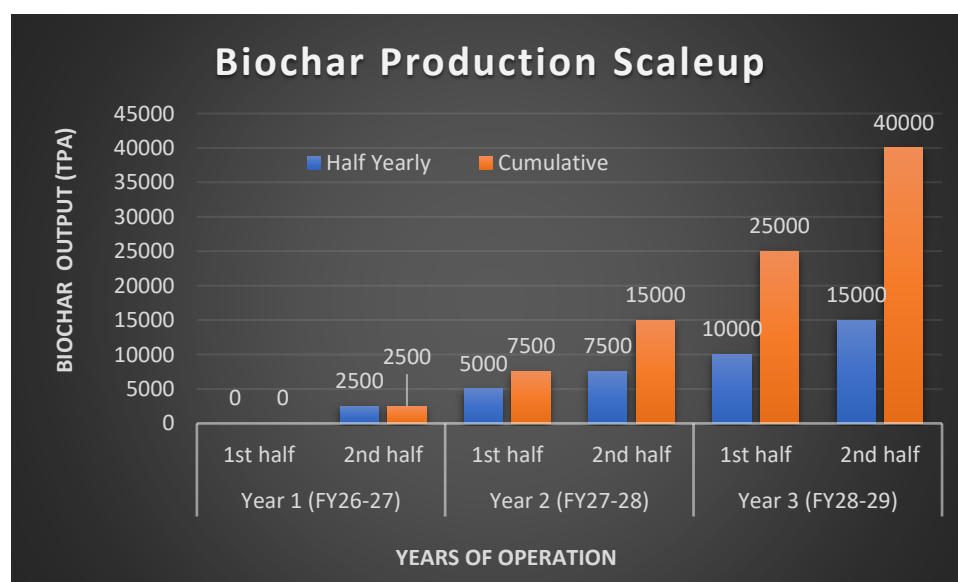
5. PROJECT OBJECTIVES & SCALABLE MODEL

A. Biochar Production scale up

Built up over 3 years, during FY 2026-27, 2027-28 & 2028-29:

- End of 6 months: 1 No 1 TPH continuous 2,500 TPA biochar capacity
- End of 12 months: add 1 No 1 TPH continuous 2,500 TPA biochar capacity
- End of 18 months: add 1 No 1 TPH continuous 2,500 TPA biochar capacity
- End of 24 months: add 1 No 1 TPH continuous 2,500 TPA biochar capacity
- End of 30 months: add 1 No 1 TPH continuous 2,500 TPA biochar capacity
- End of 36 months: add 1 No 1 TPH continuous 2,500 TPA biochar capacity

Biochar Output (year-wise) TPA						
Line ID	Year 1 (FY26-27)		Year 2 (FY27-28)		Year 3 (FY28-29)	
	1st half	2nd half	1st half	2nd half	1st half	2nd half
1	0	2500	2500	2500	2500	2500
2	0	0	2500	2500	2500	2500
3	0	0	0	2500	2500	2500
4	0	0	0	0	2500	2500
5	0	0	0	0	0	2500
6	0	0	0	0	0	2500
Half Yearly	0	2500	5000	7500	10000	15000
Cumulative	0	2500	7500	15000	25000	40000



B. Biochar Sequestration scale up

The following table outlines the growth in sequestration capacity, land impact, and carbon credit generation over the next four fiscal years.

Fiscal Year	CO2 Removal Target (tCO2)	Est. Biochar Sequestered (TPA)	Active Modular Units	Land Coverage (Hectares)
FY 2026-27	3,300	~1,650	1 (Pilot/Ramp-up)	825
FY 2027-28	13,300	~6,650	2 Units	3,325
FY 2028-29	26,600	~13,300	4 Units	6,650
FY 2029-30+	40,000	~20,000	6 Units	10,000

Execution Milestones

1. Capacity & Output (The "40k Target")

- **Steady State:** From FY 2029-30 onwards, we will maintain a consistent removal of 40,000 tCO2 per year for a 10-year period (5+5 year contract structure).
- **Biochar Volume:** This requires the sequestration of 20,000 TPA of biochar, derived from approximately 80,000 TPA of agricultural waste (coconut frond, paddy straw and cane trash).
- **Bio-Oil Contribution:** High-temperature pyrolysis will simultaneously yield bio-oil for cash flow, with the "Bio-Binder" residue contributing to the overall permanence factor and CO2 credit volume.

2. Agricultural & Farmer Integration

- **Land Management:** To achieve the 40,000 tCO2 target, we will manage 10,000 hectares of farmland.
- **Paddy & Sugarcane Strategy:** Application remains optimized at 2 tons of biochar per hectare. This cycle ensures a continuous supply of feedstock (1 ton biochar produced per 4 tons of waste collected), creating a closed-loop system with the farmers.
- **Soil Organic Carbon (SOC):** This scale-up will add approximately 15,000 tons of SOC annually across our partner farms, significantly increasing crop resilience and yield.

3. Carbon Credit Portfolio & Energy Blending

- **High-Fidelity Credits:** Based on a Permanence Factor of 0.8, our sequestration methodology ensures high-integrity carbon removal credits that meet international standards (PDD design norms).
- **Energy Synergy:** The 40,000 tCO2 removal capacity allows us to offset the carbon footprint of a significant Variable RE + BESS portfolio, providing firm, carbon-free energy to industrial clients with a high Capacity Utilization Factor (CUF).

Financial Sustainability Note

Each modular unit's feedstock procurement is de-risked by:

1. Bio-oil Sales: 12% yield of feedstock provides immediate liquidity.

Carbon Revenue: The contracted CO₂ removal (rising from 3,300 to 40,000 tCO₂) provides long-term, predictable revenue streams for 10 years.

SKILLING & GREEN ENTREPRENEURS DEVELOPMENT

6. SKILLING & GREEN ENTREPRENEURS DEVELOPMENT

As a Pilot Project of Skill Council for Green Jobs (SCGJ) for Mandya district Education & Vocational Training, will be established in KR Pet Taluka.

Have institutional tie up with 7 institutions as SCJ Training partners:

- 1 Polytechnic in KR Pet
- 6 Higher Secondary Schools in each of the 6 Hoblis

Establish a Master Training Center in Bikasandara village, Santhebachhalli hobli. This will have classroom as well as Training information covering the below topics:

1. Solar Power & Solar pump
2. Battery Energy Storage System (BESS)
3. Water Treatment
4. Small biogas plant
5. Agri residues depot management
6. Agri residues pulverized & small pelletize
7. 3D model of Electrolyser.

Conduct 8 batches of 30 students per batch (240 nos) - 2 batch in polytechnic and 1 batch in each higher secondary school.

Develop & mentor 18 to 24 entrepreneurial groups, with 120 skilled technicians which will provide 50% of the students with dignified & fulfilling livelihoods locally. Support balance 120 skilled technicians in securing green jobs or pursue higher education.

On successful implementation of Pilot, it will then be extended to 100 institutions in Mandya district, from Sept 2026 to March 2028. Creating over 1,000 local entrepreneurial jobs. This will organically grow to annually create over 5,000 local entrepreneurial jobs by 2030.

a. Agronomic Mentoring via GAM (Grameena Abhivrudhi Mandal)

GAM, a Section 8 non-profit entity, will act as the primary mentoring partner for developing local green entrepreneurs. GAM's mandate is to shift traditional practices toward Sustainable, Value-Added Farming through:

- **Skill Development:** Setting up centralised village-level / taluka-level skill development centres to train the local farmers in biomass collection, aggregation & pre-processing activities
- **Controlled Environment Agriculture (CEA):** Promoting high-nutrition and medicinal vegetables within polyhouses to maximize yield per square foot.
- **Soil Health Management:** Technical assistance in the co-application of biochar and organic amendments to restore soil biology.

b. Diversification and Market Linkages

To enhance the economic resilience of smallholder farmers, GAM will facilitate the adoption of high-value cropping patterns and agro-forestry:

- **Crop Rotation:** Transitioning to Fragrant Rice during the Kharif season and Millets or Oilseeds during the Rabi season to optimize water use and market demand.
- **Bund Agro-Forestry:** Utilizing farm boundaries for "perennial income" trees such as Bheema Bamboo.
- **End-to-End Support:** Beyond production, the project provides critical market linkages and cold-chain infrastructure to ensure farmers capture a higher share of the value chain.

c. Community Stakeholder Integration

The project moves beyond a simple "vendor-buyer" relationship by treating the local community as active stakeholders. Community members are integrated into the green business ecosystem through training in:

- **Carbon Literacy:** Understanding Soil Organic Carbon (SOC) and its link to global carbon markets.
- **Bio-Product Utilization:** Training in the application of Wood Vinegar as a bio-stimulant and the role of Bio-Binders in local infrastructure.
- **Job Creation:** Direct employment opportunities in the cold chain, logistics, and plant operations, ensuring that the wealth generated from carbon removal stays within the region.

A. Supply Chain Management

Biomass Supply Chain & Logistics Framework

The project employs a decentralized "Hub-and-Spoke" model for biomass procurement, focusing on rural entrepreneurship and rigorous quality control to ensure a steady supply of high-quality feedstock.

a. Development of Local Green Entrepreneurs (LGEs)

To ensure long-term feedstock security, the project will identify and skill **Local Green Entrepreneurs**. These stakeholders are responsible for the first-mile logistics: field-level collection, central aggregation, and primary pre-processing (chipping and powdering). By empowering local contractors, the project converts agricultural waste—traditionally a liability—into a commercial asset, fostering rural economic growth and preventing open-field burning.

b. Logistical Optimization and Carbon Leakage Monitoring

Feedstock transport is optimized based on the distance from the aggregation point to the modular plant:

- **Short-Haul (< 20 km):** Tractor-trolleys will be utilized for cost-efficiency and manoeuvrability in rural terrain.
- **Long-Haul (> 20 – 50 km):** Standardized trucks will be deployed for higher volume delivery. Fuel consumption for every consignment is recorded to compute transportation emissions & carbon leakage monitoring.

c. Procurement Strategy and Contractual Obligations

To mitigate supply chain risks, the project will establish a diversified supplier base. The goal is to identify at least 10 primary suppliers, each capable of delivering a

minimum of 10 tons of biomass per day. This diversification ensures that the modular plant maintains its 24/7 operational cycle even if individual supply nodes face seasonal disruptions.

d. Quality Assurance and Material Receipt System (MRS)

A robust Material Receipt System will be installed at the plant biomass storage area to ensure that only feedstock meeting the modular plant design specifications enters the system.

- **Primary Verification:** All consignments undergo weighing on a calibrated In-house Weigh Bridge.
- **Sampling and Moisture Control:** Random Sampling will be carried out to ensure data accuracy, samples will be collected from every consignment coming in with the pre-processed biomass for moisture verification.
 - **Moisture Threshold:** Feedstock must maintain a moisture content below **10%–12%** to ensure optimal pyrolysis efficiency and prevent energy loss during the drying phase.
- **Laboratory Characterization:** For every consignment received, two representative samples are sent to the on-site laboratory for proximate and ultimate analysis (specifically monitoring ash content and calorific value). This ensures the 72%–75% carbon content of the final biochar remains consistent.

***GREEN
ENTREPRENEURS
MENTORING &
OPERATIONS
MANAGEMENT***

7. GREEN ENTREPRENEURS MENTORING & OPERATIONS MANAGEMENT

7.1 GAM Mentoring Structure for Local Green Entrepreneurs (LGEs)

Grameena Abhivrudhi Mandali (GAM) is the designated field mentoring partner for converting the 240 SCGJ-trained candidates (Section 10) into active Local Green Entrepreneurs (LGEs). An LGE is a trained individual or small group (2–5 persons) who independently operates one or more biomass collection, aggregation, and pre-processing nodes within their home village or cluster, under a formal Service Agreement with Shree Krishna CDR. Each LGE manages a defined catchment of approximately 500–1,000 cultivated acres and is responsible for delivering a minimum of 5 tonnes of dry, process-ready biomass per week to the nearest Village Collection Centre or directly to the plant. GAM assigns one Field Officer per 6 LGEs, conducting fortnightly farm visits, equipment checks, and MRV data audits during Year 1, transitioning to monthly visits from Year 2 as LGEs stabilise.

7.2 Equipment Leasing and Revenue Model

Shree Krishna CDR procures mobile chippers and balers as project assets; LGEs operate equipment under a daily hire arrangement (estimated Rs 800–1,200 per day for a chipper including fuel) charged against their monthly earnings. LGE earnings are settled on a dry-weight, moisture-verified basis, ensuring payment for quality rather than gross volume. LGEs also receive a cash component from the barter + cash split procurement model: for every tonne of dry biomass delivered, the LGE earns a logistics margin over and above the farmgate barter value. After 24 months of consistent performance, high-performing LGEs are eligible for an Equipment Purchase Loan at concessional rates (with GAM as guarantor) under PM MUDRA or NABARD agri-value-chain schemes, enabling ownership and improved margins.

7.3 Performance Metrics and Escalation

GAM field officers conduct monthly performance reviews against three KPIs:

- (1) Volume delivered — minimum 20 T/month dry basis per LGE to qualify for bonus tier
- (2) Moisture compliance — minimum 85% of deliveries must meet the $\leq 20\%$ moisture specification, with rejections above 15% triggering a mentoring visit
- (3) MRV data quality — GPS tags, weight records, and moisture readings uploaded within 24 hours of each delivery.

Tier 1 LGEs (meeting all KPIs for 6 consecutive months) receive extended catchment areas and priority equipment access. LGEs underperforming for two consecutive months receive a structured improvement plan. After three consecutive months of non-compliance, the catchment area is reassigned to a substitute from the SCGJ reserve pool.

Escalation path: LGE → GAM Field Officer → Shree Krishna CDR Supply Chain Manager, with a 48-hour resolution SLA at each tier.

PROJECT COMMAND AREA

8. PROJECT COMMAND AREA

Taluka-Wise Dominant Crop Patterns

Taluka	Dominant Cropping Pattern	Irrigation Profile
K R Pet	Paddy + Sugarcane + Dairy	Highly irrigated by Hemavathi and KRS canals.
Pandavapura	Sugarcane + Paddy + Horticulture	Central part of the sugar belt; heavy reliance on KRS canals.
Channarayapatna*	Maize + Ragi + Coconut	Shift toward commercial maize and plantation crops; mixed irrigation.
Nagamangala	Ragi + Pulses + Livestock	Predominantly rain-fed; higher focus on millets and fodder.

Taluka	District	Net Sown Area (Ha)	Primary Crop (Acreage)	Secondary Crop (Acreage)	Annual Surplus Biomass (Tonnes)	Dominant Feedstock Type
K R Pet	Mandya	~48,500	Coconut (~13,500)	Paddy (~10,200)	~110,000	Coconut Fronds & Paddy Straw
Pandavapura	Mandya	~34,000	Paddy (~11,800)	Sugarcane (~9,100)	~165,000	Cane Trash & Paddy Straw
Nagamangala	Mandya	~42,000	Ragi (~14,500)	Maize (~6,800)	~65,000	Ragi Straw & Maize Cobs
Channarayap atna	Hassan	~66,500	Maize (~22,500)	Paddy (~15,200)	~190,000	Maize Cobs & Paddy Straw

Land Use & Crop Distribution in K R Pet

K R Pet (Krishnarajapet) is one of the most agriculturally significant taluks in Mandya. The land use pattern is primarily driven by the **Hemavathi** and **KRS** canal networks.

- **Total Geographical Area:** ~91,000 Hectares.
- **Net Sown Area:** Approximately 45,000–50,000 Hectares.
- **Coconut Cultivation Area:** As of 2024-2025, Mandya district has ~74,731 hectares under coconut. **K R Pet is one of the top three taluks** in the district

for coconut density, with an estimated area of **12,300 to 14,000 Hectares** dedicated to coconut plantations.

- **Cropping Intensity:** High, often featuring intercropping with Banana, Cocoa, or Drumstick.

2. Feedstock Availability: Coconut Fronds

Coconut fronds (leaves) are an ideal "low-lignin, high-carbon" feedstock for modular pyrolysis units. They are often treated as "plantation waste" and are currently underutilized or burnt on-site.

A. Biomass Generation Statistics

Based on standard agronomic yields for the *West Coast Tall* and *Hybrid* varieties dominant in K R Pet:

- **Average Frond Fall:** 12–15 fronds per tree per year.
- **Plantation Density:** ~70 trees per acre (approx. 175 trees per hectare).
- **Weight per Frond (Dry):** ~1.5 kg to 2.0 kg per frond.

B. Estimation Table (K R Pet Taluk)

Parameter	Value (Estimated)
Total Coconut Area (K R Pet)	12,500 Hectares
Total Tree Population	~2.1 Million Trees
Annual Frond Generation	~28 Million Fronds
Total Dry Biomass Potential	~42,000 – 56,000 Tonnes/Year
Sourcing Accessibility	High (clustered along canal banks)

3. Feedstock Suitability for Pyrolysis

Coconut fronds possess unique thermochemical properties that make them superior to other agri-residues:

- **Calorific Value:** ~17.5 MJ/kg (high energy content for self-sustaining pyrolysis).
- **Carbon Content:** High fixed carbon (~17-20% in raw state), yielding high-density biochar.
- **Physical Structure:** Large surface area when shredded, allowing for rapid and uniform heat transfer in the pyrolyzer drum.

4. Supply Chain Management (SCM) Strategy

For the **SHREE KRISHNA CDR** business unit, the Sourcing & Logistics plan should follow this localized model:

1. **Collection (The "Pull" System):** Establish collection points at the village level. Farmers in K R Pet typically prune trees twice a year (Pre-monsoon and Post-monsoon), creating "peak" supply windows.
2. **On-site Shredding:** Raw fronds are bulky and expensive to transport. Mobile shredders should be deployed to reduce the volume by 4:1 before moving to the modular plant.
3. **Moisture Management:** Fronds in Mandya have a moisture content of 40–50% when fresh. A decentralized "sun-drying" yard at the collection hub can reduce this to <20% to save energy in the pyrolyzer.
4. **Logistics Cost:** Given the high density of plantations in K R Pet, a **15 km sourcing radius** can typically supply a 1-tonne/hour modular plant at optimal transport costs.

Reference Note: "All crop acreage and district-level statistics are cross-referenced with the Mandya District Agriculture Contingency Plan and the Horticulture Department's Annual Action Plan 2025-26 to ensure alignment with SEBI's BRSR Value Chain reporting requirements."

Land Use and Crop Distribution (Pandavapura)

Pandavapura's land use is almost entirely defined by the **Visveswaraya Canal** (from the KRS dam). Unlike Nagamangala, which is rain-fed, Pandavapura is characterized by intensive, water-heavy monocropping.

- **Total Cultivated Area:** Approximately 32,000–35,000 Hectares.
- **Sugarcane Area (2024-25):** Approximately **8,500 – 9,500 Hectares**. It is the primary catchment area for the **Pandavapura Sahakari Sakkare Karkhane (PSSK)**.
- **Paddy Area (2024-25):** Approximately **11,000 – 12,500 Hectares**.
- **Irrigation Profile:** >90% of the net sown area is irrigated, ensuring consistent biomass yields even in erratic monsoons.

Biomass Availability Estimates

A. Sugarcane Trash (Cane Trash)

Cane trash consists of the green leaves and dry tops left in the field after harvesting the stalks for the mill.

- **Trash Generation Rate:** ~140 kg of dry trash per tonne of sugarcane stalk (or ~10–12 tonnes per hectare).
- **Estimated Annual Surplus:** ~85,000 – 110,000 Tonnes.
- **Sourcing Advantage:** Often burnt by farmers to clear fields for the next ratoon crop; high "willingness to supply" for biochar units.

B. Paddy Straw

Paddy straw is the byproduct of grain harvesting. In Pandavapura, the prevalence of mechanical harvesters has increased the availability of "loose straw" which is difficult for fodder use.

- **Straw-to-Grain Ratio:** ~1.3 : 1.
- **Estimated Annual Surplus:** ~65,000 – 75,000 Tonnes.
- **Sourcing Advantage:** High availability in two distinct seasons (Kharif and Summer/Rabi).

Feedstock Potential for Biochar Production

Feedstock	Carbon Content (Dry)	Calorific Value	Pyrolysis Yield (Biochar)
Cane Trash	42–45%	~17.8 MJ/kg	25–30%
Paddy Straw	38–40%	~15.0 MJ/kg	20–25%

Supply Chain Reference Data (Mandya Cluster)

For the DPR, the following logistics parameters are specific to the Pandavapura-Srirangapatna belt:

- **Collection Radius:** A 10 km radius in Pandavapura provides nearly 15,000 tonnes of mixed biomass, enough to fuel a 2 TPH (Tonnes Per Hour) modular plant.
- **Logistics Seasonality:** * **Paddy:** Peak availability in Nov–Dec and May–June.
 - **Sugarcane:** Continuous availability from Nov to May (matching mill crushing seasons).

Data Source References

The statistics and land use patterns provided in this report are sourced from:

1. **Mandya District Agricultural Progress Report (Dec 2024):** Published by the Deputy Commissioner of Mandya, citing district-wide sowing of 57,084 Ha (Paddy) and 40,361 Ha (Sugarcane).

2. **Chinimandi / The Hindu (2024-25 Agri Reviews):** Reports on the 2024-25 agricultural season and PM Fasal Bima enrollment for Pandavapura Taluk.
3. **District Irrigation Plan (PMKSY):** Section 5.5 "Strategic Action Plan for Pandavapura Taluk" which details land use, soil pH, and cropping intensity.
4. **ICAR-IISc Technical Report (TR109):** "Bioresource Potential of Karnataka" providing biomass-to-demand ratios for the Southern Dry Zone.
5. **Journal of Sugarcane Research (2015/2024):** Studies on jaggery and sugar mill catchment areas in Pandavapura and K.R. Pet.

Land Use and Crop Distribution (Channarayapatna)

Channarayapatna is a high-productivity taluk where agriculture is a mix of intensive canal-irrigated crops and semi-dry commercial crops. It serves as a major gateway for agricultural trade in the region.

- **Total Cultivated Area:** ~65,000 – 68,000 Hectares.
- **Paddy Area (2024-25):** Approximately **14,000 – 16,500 Hectares**. Paddy is concentrated in the southern and eastern parts of the taluk, benefiting from the Hemavathi canal network.
- **Sugarcane Area (2024-25):** Approximately **4,200 – 5,500 Hectares**. While lower than Pandavapura, it is a significant commercial crop that feeds the **Chamundeshwari Sugars** and other nearby crushing units.
- **Maize (The "Competitor"):** Notably, maize has expanded significantly here (~22,000 Ha), often competing with paddy for land. For the CDR unit, maize cobs represent a secondary biomass stream.

2. Biomass Availability Estimates

A. Paddy Straw

With two primary growing seasons (Kharif and Summer), Channarayapatna produces a massive surplus of straw.

- **Straw Generation Rate:** ~5.5 to 6.5 tonnes per hectare (dry weight).
- **Estimated Annual Surplus:** ~80,000 – 105,000 Tonnes.
- **Current Usage:** Unlike Nagamangala, where straw is saved for fodder, the mechanical harvesting in Channarayapatna leads to high volumes of "stubble" often burnt to facilitate quick turnaround for the next crop.

B. Sugarcane Trash (Cane Trash)

- **Trash Generation Rate:** ~10 tonnes per hectare.

- **Estimated Annual Surplus: ~42,000 – 55,000 Tonnes.**
- **Sourcing Advantage:** The trash is often available in concentrated clusters near the borders of K R Pet and Srirangapatna, simplifying transport logistics for a modular plant.

3. Comparative Biomass Potential

Parameter	Paddy Straw (Pandavapura)	Paddy Straw (Channarayapatna)
Acreage	~12,000 Ha	~15,000 Ha
Biomass Yield	High (Canal Fed)	High (Mixed/Canal Fed)
Key Residue	Fine Straw	Coarse Straw + Maize Cobs
Primary Use	Some Fodder / Burning	Heavy Burning / Commercial Waste

4. Supply Chain Sourcing Strategy

For the **Shree Krishna CDR** unit, KR Pet offers a unique "Dual-Residue" sourcing model:

1. **Direct Farmer Offtake:** Since Channarayapatna has a strong network of FPOs (Farmer Producer Organizations), the supply chain can be managed through **Village Collection Centers (VCCs)**.
2. **Maize Integration:** In months when paddy straw is scarce (July-August), the taluk's massive maize production provides "Maize Cobs" and "Stalks," which can be processed in the same modular pyrolyzer with minor adjustments to the feed screw.

5. Data Source Reference

The statistics and land use data for Channarayapatna are sourced from:

1. **Hassan District Socio-Economic Review (2024-25):** Statistical data on taluk-wise crop coverage provided by the *Directorate of Economics and Statistics, Govt. of Karnataka*.
2. **Joint Director of Agriculture (JDA) Hassan - Annual Progress Report (Nov 2024):** Preliminary reports on crop damage and actual sown area for Channarayapatna Taluk.
3. **Karnataka District Factbook™ (Hassan District - Jan 2026 Edition):** Detailed demographics and total cropped area (5,74,050 Ha for the district) with taluk-level breakdowns.
4. **ICAR-IIISc Technical Report (Bioresource Potential):** Standardized biomass-to-grain ratios for Hassan district's agro-climatic zones.
5. **Department of Horticulture, Hassan:** Survey data on perennial crops (Coconut) that often border paddy fields in the taluk.

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Water Assessment

Based on the 2022 assessment by the Central Ground Water Board (CGWB) and associated studies, the groundwater situation in the Mandya district is characterized by a stage of extraction of 62%, indicating that the district is in the "Safe" category, though certain taluks face higher stress.

Groundwater Availability and Usage (2022 Data):

- **Total Annual Ground Water Recharge:** 71,194.93 Hectare metres (Ham).
- **Annual Extractable Ground Water Resource:** 64,599.53 Ham.
- **Annual Ground Water Extraction:** 40,193.83 Ham.
- **Stage of Extraction:** 62%.
- **Future Availability:** There is significant, albeit varying, availability for future irrigation in the district.

Taluk-Level Variations:

- **Pandavapura Taluk:** The annual extractable water resource is 8,125.09 Ham, with a net groundwater availability for future use of 3,393.99 Ham.
- **Maddur Taluk:** The annual extractable water resource is 8,154.38 Ham, with a net availability for future use of 3,625.49 Ham.
- **Nagamangala Taluk:** Studies indicate that 85% of the area has low, and the rest has moderate, groundwater recharge potentiality.

Water Levels and Quality:

- **Water Depth:** During the pre-monsoon period (May 2019), water levels in aquifers varied, with many areas having levels between 5 to 10 meters below ground level (mbgl).
- **Seasonal Fluctuation:** The region experiences significant fluctuations, with 55% of wells in the state showing a fall in water level, and 45% showing a rise as of late 2020.
- **Quality Issues:** In Mandya City, studies indicate that 60% of water samples were of poor quality, largely due to high levels of iron and fluoride.

- **Contamination:** Higher radon concentrations (a natural radioactive gas) were observed in borewell water, particularly in Mandya city and Yettaganahalli.

Hydrogeology:

- Groundwater primarily occurs in weathered formations like granitic gneiss, granite, and schists, with depths up to 25 m (shallow) and 200 m (deep).
- The soil is generally highly leached with low water holding capacity

Source: <https://link.springer.com/article/10.1007/s40899-023-00998-w#citeas>

***FEEDSTOCK SUPPLY
CHAIN MANAGEMENT
- AT SOURCE***

9. FEEDSTOCK SUPPLY CHAIN MANAGEMENT - AT SOURCE

"Month-wise Collection Target" (in Tonnes) for each taluka

Operating a 2-line, 1-TPH modular plant on a 24/6 schedule (24 hours a day, 6 days a week) requires well planned logistics. With two lines, the hourly intake is 2 tonnes, amounting to 48 tonnes per day and approximately 1,150–1,200 tonnes per month.

24/6 Operational Collection Target (Tonnes per Month)

Total Monthly Intake Required: ~1,200 Tonnes

Month	K R Pet (Coconut & Paddy)	Pandavapura (Cane & Paddy)	Nagamangala (Paddy & Ragi*)	Channarayapatna (Paddy Straw)	Monthly Total (Tonnes)	Primary Feedstock Strategy
Jan	250	500	100	350	1,200	Cane Trash Peak: Focus on Pandavapura.
Feb	250	500	100	350	1,200	Cane Trash Peak: Focus on Pandavapura.
Mar	500	400	50	250	1,200	Coconut Pruning: Maximize K R Pet.
Apr	500	400	50	250	1,200	Dry Stocking: High- density fronds.
May	300	200	200	500	1,200	Summer Paddy: Intake from Channarayapatna.
Jun	200	200	300	500	1,200	Summer Paddy: Maximize Channarayapatna.
Jul	400	300	250	250	1,200	Lean Month: Utilizing Coconut buffers.
Aug	400	300	250	250	1,200	Monsoon Reserve: Focus on stored fronds.
Sep	400	400	100	300	1,200	Stabilizing: Mixing dry fronds with straw.
Oct	300	400	200	300	1,200	Inventory Prep: Clearing yards for Kharif.

Nov	200	450	100	450	1,200	Kharif Peak: High-volume Paddy Straw.
Dec	200	500	100	400	1,200	Kharif Peak: Intensive collection.
Annual	3,900	4,600	1,750	4,150	14,400	

"Year-wise Collection Target" (in Tonnes) for each taluka

To secure a net supply of 10,000 TPA (Tonnes Per Annum) of process-ready biomass, we must account for the Moisture Content (MC) and Process Losses (handling, storage, and transport dust/debris).

In agricultural residues, "As-Received" biomass often contains 30–60% moisture. Since the pyrolysis lines require dry feedstock (typically <15–20% MC), it is necessary to collect a higher gross weight to net the required dry tonnage.

1. Calculation Logic for 10,000 TPA Net Goal

- Target Dry Feedstock: 10,000 Tonnes.
- Moisture Loss (Evaporation): ~25% (Average weight loss from field-moist to process-ready dry).
- Process/Handling Loss: ~5% (Shattering, dust, and non-combustible debris).
- Gross Collection Target: ~14,300 Tonnes of Raw Biomass.

2. Village Clusters & Feedstock Breakup (Accounting for Losses)

This table lists the villages with the highest density of these specific crops to minimize the "transport-per-tonne" cost.

Taluka	Target Village Clusters	Gross Raw Supply (MT)	Net Dry Supply (MT)	Feedstock Composition
K R Pet	Akkihebbal, Kikkeri, Bookanakere	4,000	2,800	Coconut Fronds: Naturally drier, stable supply.
Pandavapura	Chinakurali, Harohalli, Kyathanahalli	4,500	3,150	Sugarcane Trash (High MC): Requires sun-drying.

Nagamangala	Belluru, Bindiganavile, Honakere	2,000	1,400	Ragi Straw/Pulses: Mixed dry residues.
Channarayapatna	Bagur, Janivara, Shravanabelagola	3,800	2,660	Paddy Straw: High volume, seasonal peaks.
TOTAL		14,300	10,010	

3. Moisture & Loss Profile by Feedstock

Different materials behave differently during the "curing" (drying) phase at the Village Collection Centers (VCCs).

Feedstock	Field Moisture (%)	Target Process Moisture Content (%)	Weight Loss Factor	Storage Risk
Coconut Fronds	20% – 30%	15%	~10%	Low (Very stable).
Paddy Straw	25% – 35%	12%	~20%	High (Fire risk when dry).
Sugarcane Trash	45% – 55%	15%	~40%	High (Internal heating / mold if wet).

4. Implementation Strategy for the Village Clusters

- A. K R Pet (Kikkeri Cluster): This is the most efficient supply chain. Coconut fronds have the lowest "Process Loss" because they are bulky and don't shatter like straw. They should be the "base load" during the monsoon.
- B. Pandavapura (Chinakurali Cluster): sugarcane trash has the highest moisture loss (up to 40%), the VCC must have extended drying yards. Weighing / buying must be based on dry-matter weight or after 48 hours of field-drying.
- C. Channarayapatna (Bagur Cluster): Paddy straw "shattering" (process loss) is high. Ensure the balers are well-maintained to keep the straw tightly packed, otherwise, there will be a loss of 5–7% of weight as "dust" during transport.

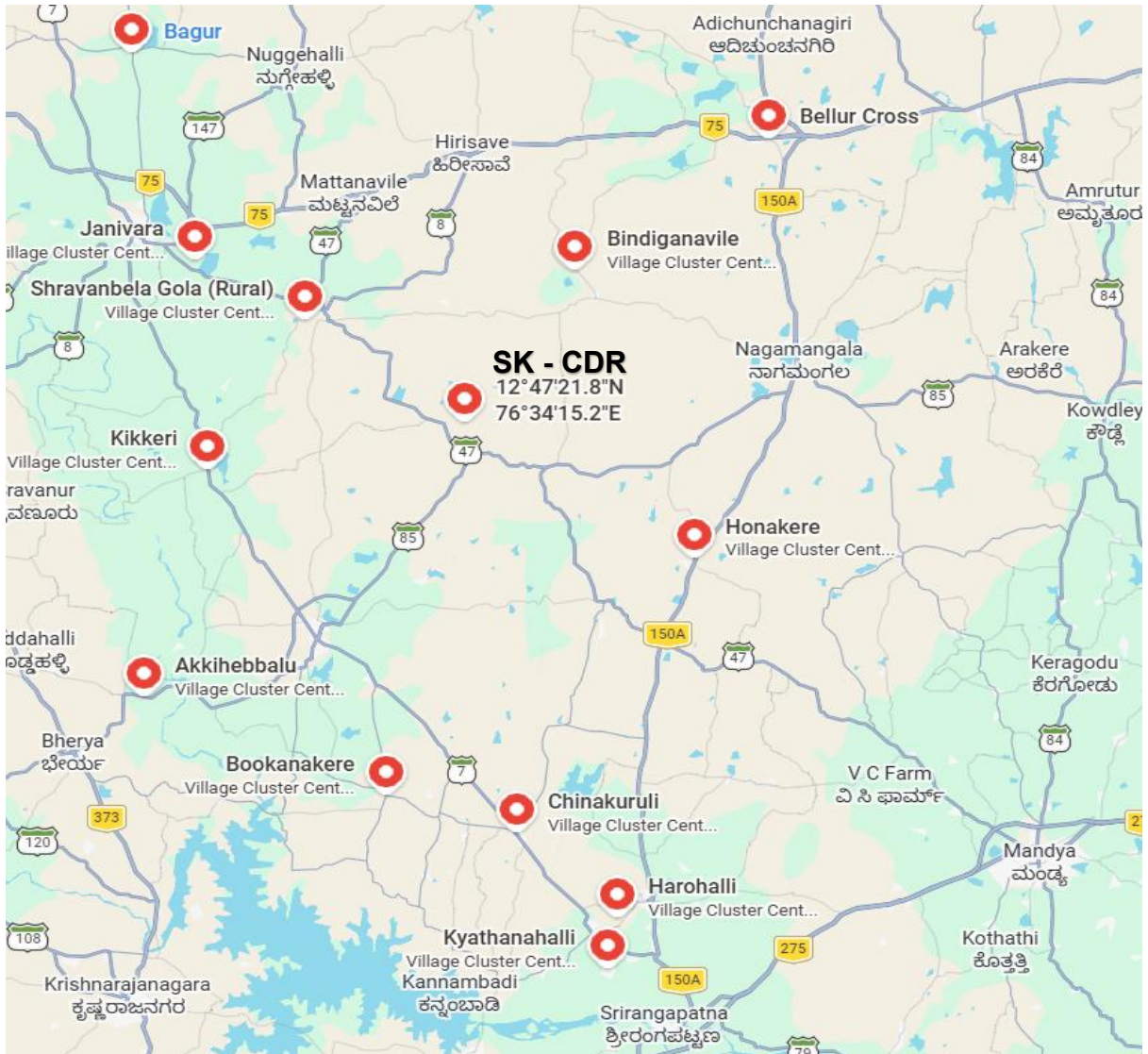
Data Source Reference

- Moisture Data: ICAR-Central Institute of Agricultural Engineering (CIAE) - Post-Harvest Technology Handbook.

- Village Cropping Intensity: *Karnataka State Remote Sensing Applications Centre (KSRSAC) - Land Use/Land Cover Maps 2024*.
- Process Yields: *Biochar Quality Mandate (IBI Standards) & Pyrolysis Process Engineering Data (Slow Pyrolysis)*.

To visualize the supply chain, the village clusters are mapped relative to the plant location in K R Pet (Latitude: 12°47'21.3"N, Longitude: 76°34'15.5"E). This "Hub-and-Spoke" map shows that most of the key sourcing points are within a 20-35 km radius, which is ideal for minimizing logistics costs and carbon emissions.

Modular Biomass Plant Cluster Map



Taluka	Village Cluster	Approx. Distance to Plant	Feedstock Role
K R Pet	Bookanakere	~6 km	Strategic Hub: Nearest year-round Coconut supply.
K R Pet	Akkihebbalu	~19 km	Major Coconut Frond sourcing zone.
Pandavapura	Chinakuruli	~9 km	Primary Hub: Highest density Sugarcane Trash.
Pandavapura	Harohalli	~15 km	High-volume Paddy Straw & Sugarcane.
Nagamangala	Honakere	~29 km	Ragi Straw & Pulse Residue for dry season.
Hassan	Shravanbela Gola	~38 km	Outer Rim: Secondary Paddy Straw source.

Logistics Optimization Insights

Bookanakere

- The "Golden Circle" (0-15 km): Bookanakere, Chinakuruli and Kyathanahalli fall within the immediate vicinity. These villages should provide roughly 50% of the daily 48-tonne requirement to keep transport costs at a minimum.

Janivara

- High-Yield Clusters: Janivara and Bagur in Channarayapatna are slightly further out (~45-55 km) but offer massive paddy straw volumes. For these distances, high-density baling is mandatory to ensure each truck trip is financially viable.

Bindiganavile

- Monsoon Stability: During the heavy rains, prioritize Bindiganavile and Honakere. These areas typically harvest Ragi and pulses, which have lower moisture content and better storage stability than canal-irrigated straw.

Recommended Feedstock Routing

1. Direct Feed: Daily tractor-trailers from Pandavapura and K R Pet clusters.
2. Staging Hubs: Use Village Collection Centers (VCCs) in Kikkeri and Bellur to aggregate and dry material before the final leg to the plant.

Feedstock sourcing phase-wise scaleup plan

To accommodate a massive scaling plan—from a **2 TPH** (Phase 1) to a **12 TPH** (Phase 6) facility - the sourcing strategy must shift from local village collection to a regional industrial supply chain.

For the annual requirement at Phase 6 (if running 24/6) will eventually exceed **85,000 TPA**. However, for the initial target of **16,000 TPA** (Phase 1-3), the strategy focuses heavily on the high-density Coconut belt of K R Pet.

1. Phase 1–3: The K R Pet Monopoly (16,000 TPA)

In these phases, you are operating from one building (2 lines). K R Pet is the "anchor" due to its year-round coconut frond availability.

Feedstock Allocation & Sourcing

- **Coconut Fronds (8,000 TPA):** This requires tapping into approximately **4,000–5,000 hectares** of coconut plantations.
 - **Hubs:** Focus on **Akkihebbalu** and **Bookanakere**. These are "High-Density Frond Zones."
 - **Automation:** Deploy **Mobile Shredders** at the plantation gate. Coconut fronds are 70% air; shredding at the source increases transport efficiency by 300%.
- **Paddy & Cane (8,000 TPA):**
 - **Paddy Straw:** Sourced from the Hemavathi canal command areas of K R Pet during Dec–Jan and May–June.
 - **Cane Trash:** Sourced from the catchment area of the Hemavathi Sahakari Sakkare Karkhane (HSSK).

2. Phase 4–6: Regional Expansion (Scaling to 12 TPH)

As the plant scales to 6 buildings, K R Pet alone will not sustain the volume without causing price inflation. The strategy must expand to **Pandavapura, Channarayapatna, and Nagamangala**.

Sourcing Multi-Taluka Strategy

- **Pandavapura (The Sugar Hub):** Becomes the primary source for **Cane Trash** to balance the 8,000 TPA requirement.
- **Channarayapatna (The Paddy Hub):** Becomes the "Volume Buffer." When K R Pet paddy straw is diverted for fodder, the mechanical-harvested surplus from Channarayapatna fills the gap.
- **Nagamangala (The Seasonal Buffer):** Used for pulse residues and dry-land stalks to maintain the 24/6 operation during heavy monsoon months when canal-belt straw is too wet.

3. Civil & Structural Scaling Requirements (5-Acre Site)

The 5-acre leased land must be designed with "Modular Expansion" in mind to accommodate the jump from 2 lines to 12 lines.

Pre-Processing Zone (The "Engine Room")

- **Structure:** A central **Processing Building** (PEB) housing 3 sets of heavy-duty industrial lines.
- **Equipment:** *
 - **Primary Chippers:** For Coconut Fronds (Handling 30–40 TPD).
 - **Fine Grinders/Pulverizers:** To create a uniform 2-5mm feedstock mix.
 - **Structural Need:** Independent RCC foundations for each grinder to prevent harmonic vibration transfer across the building.

The "Dual-Stream" Storage Strategy

To manage 16,000 TPA growing into much larger volumes, storage must be split:

1. **Green Zone (Coconut Fronds):** Open-sided, ventilated sheds. Fronds need airflow to prevent fungal growth.
2. **Dry Zone (Straw & Processed Powder):** Fully enclosed, fire-rated silos or sheds.

MRV & Logistics Infrastructure

- **Digital Weighbridge:** Integrated with a cloud-based ERP. As the plant scales to Phase 6, the truck/tractor will be entering every 15 minutes. The civil design must include a **Loop Road** (One-way entry/exit) to prevent bottlenecks.
- **Bio-Oil Depot:** An RCC-bunded tank farm with a capacity for **15 days of production**. Includes a decantation area with oil-separators for environmental compliance.

Summary of Phase-wise Capacity

Phase	Buildings	Total TPH	Annual TPA	Primary Source	Infrastructure Focus
1 - 3	1	2 TPH	16,000	K R Pet	Primary Shredders & 1st PEB Shed.
4 - 6	3	6 TPH	~43,000	K R Pet + Pandavapura	Automated Conveyors & Buffer Silos.
Final	6	12 TPH	~86,000	Regional Cluster	Full ERP-SCM & Fire Ring Main.

TRANSPORT & LOGISTICS

10. TRANSPORT & LOGISTICS

An efficient, resilient, and transparent supply chain is fundamental to the long-term success, carbon integrity, and bankability of this model Biochar Project. Unlike conventional industrial facilities, a biochar carbon removal project depends critically on:

- Sustainable agricultural residue sourcing
- Reliable logistics under rural conditions
- Accurate material traceability from source to final product
- Human resource systems that ensure compliance, safety, and dignity

This section describes in detail the **end-to-end supply chain and logistics framework**, covering:

- Upstream biomass sourcing and aggregation
- Inbound logistics and material handling
- Production-interface coordination
- Outbound biochar distribution

The Supply chain has been designed based on the following principles:

1. **Localization:** Maximum sourcing within K R Pet taluka to minimize transport emissions and leakage.
2. **Diversification:** Multiple feedstock types and villages to reduce dependency risk.
3. **Traceability:** Village-level, batch-level, and product-level tracking integrated with MRV.
4. **Human-centric operations:** Creation of dignified rural employment by training and employing skilled local workforce.

The supply chain is organized into **three interlinked segments**:

1. **Upstream Supply Chain** – Biomass sourcing, aggregation and pre-processing
2. **Production-Interface Supply Chain** – Inbound logistics, processing and plant coordination
3. **Downstream Supply Chain** – Biochar storage and distribution

1. Upstream Supply Chain – Biomass Sourcing, Aggregation and pre-processing

Feedstock Types Handled

- Coconut fronds
- Sugarcane trash
- Paddy trash
- Bagasse (from mills / jaggery units)
- Bamboo residues

Biomass is sourced from approximately 90 villages within K R Pet taluka through decentralized aggregation points considering Field-level collection and bundling, temporary yards in the fields & basic preprocessing where applicable (shredding / chipping / baling). This model reduces farmer transport burden, improves feedstock quality & enhances traceability.

2. Production-Interface Supply Chain Management– Inbound logistics and plant coordination

Inbound logistics - for material handling tractor-trailers are used for short distances (<20 km), Light commercial vehicles (LCVs) for long distances (>20 kms~50 km) will be used for collection & transportation of the feedstock from the source to the plant.

Plant coordination - this segment ensures seamless coordination between inbound logistics and plant operations.

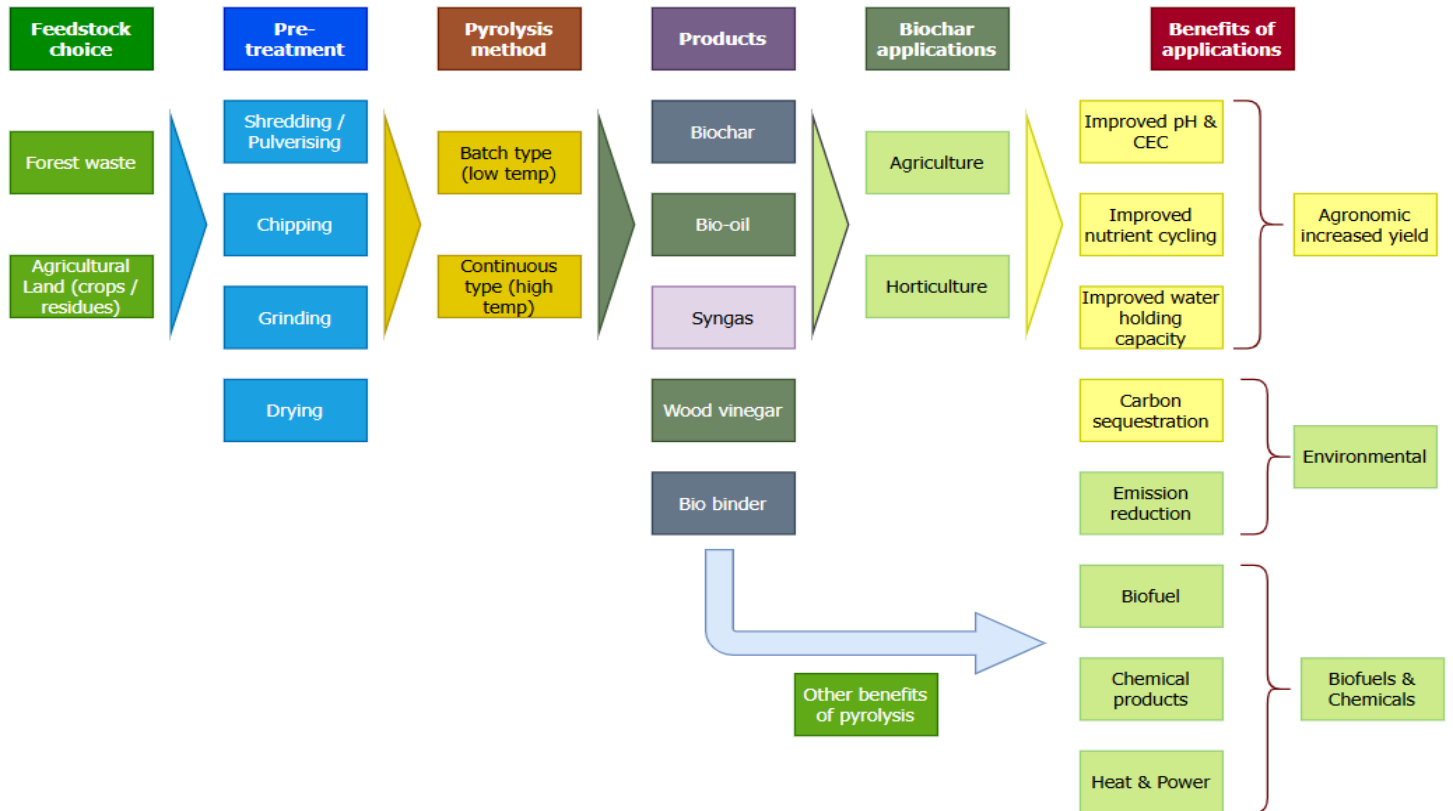
- a. Biomass Reception at Plant - weighbridge recording, moisture and visual quality checks & Assignment of batch ID
- b. Storage & Inventory Management - covered biomass yards, FIFO inventory system & feedstock segregation by type
- c. Interface with Production Planning - daily feedstock requirement forecasts, buffer stock maintenance (~30 days), coordination between logistics and operations teams

3. Downstream Supply Chain – Biochar Storage & Distribution

- a. Finished Biochar handling involves cooling and screening the biochar, bagging in 50-kg PP bags with barcode / QR code printing for traceability & stored in the finished goods godown in a designated location.
- b. The finished biochar will be supplied to farmers for soil amendment & SOC improvements.
- c. The finished biochar will be dispatched and delivered as per order-based dispatch planning with proper documentation and invoicing

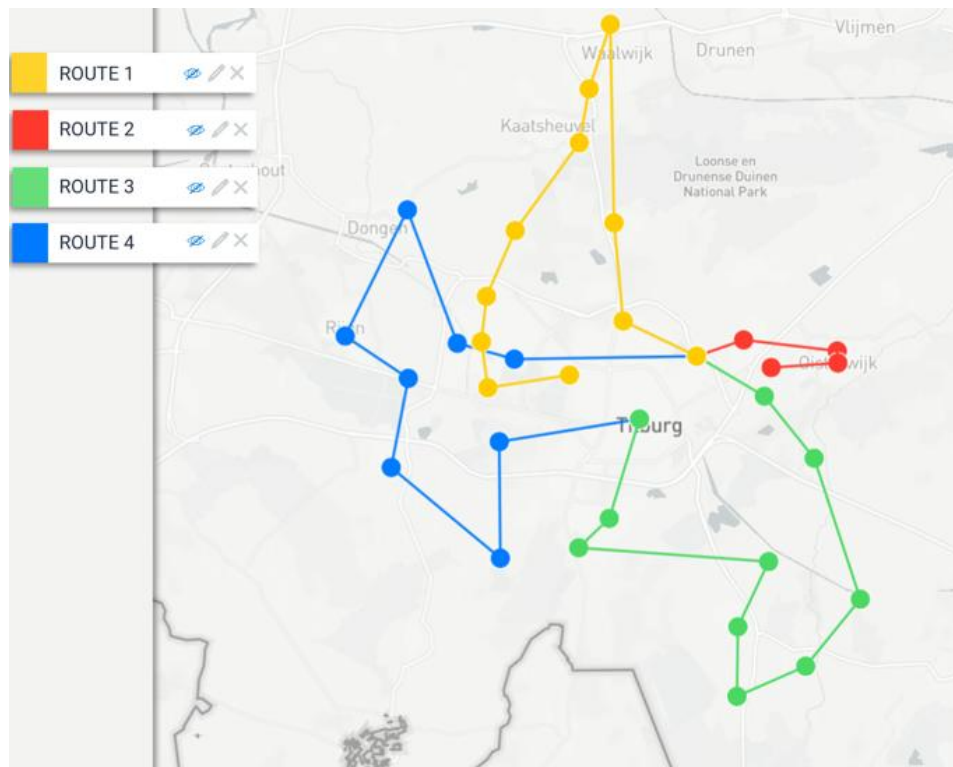
The transport & logistics is designed for Scalability and Replicability, it is modular replicable across talukas, scalable with additional pyrolysis lines. This system provides a **strong operational backbone** for sustained biochar production.

End-to-end Biochar Supply Chain Flow

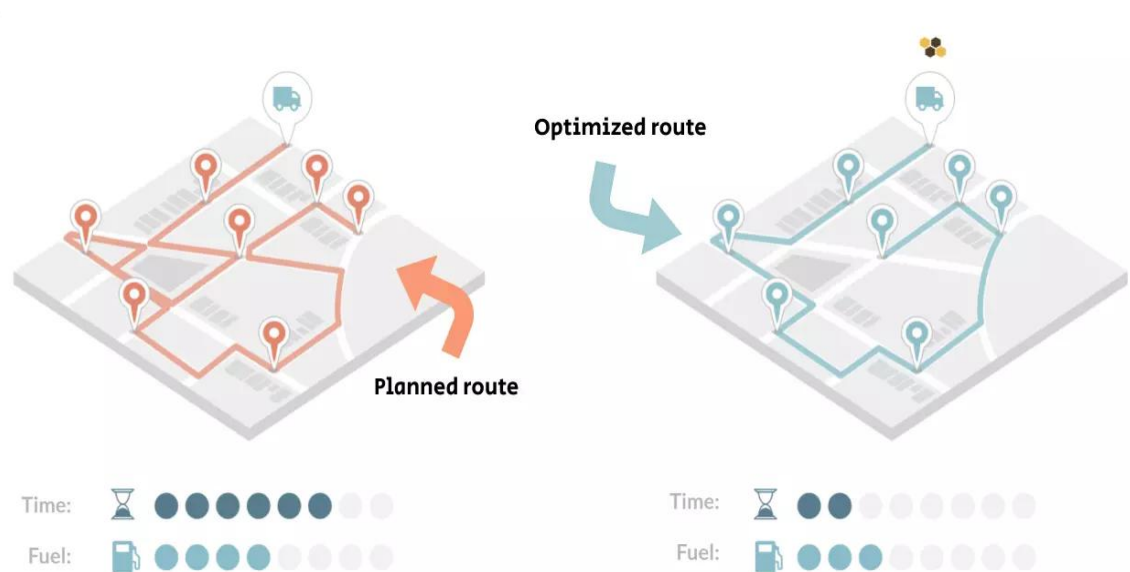


Route Optimization:

- (1) Short-haul <20 km — tractor-trailer milk-runs from Bookanakere & Chinakurali at 06:00 and 13:00 daily
- (2) Long-haul 20–50 km — LCV pre-booked slots to Harohalli, Bagur, Shravanabelagola
- (3) Monsoon routing June–Aug — prioritise coconut clusters on tarmac roads. All vehicles GPS-tracked. Target cost: Rs 800–1,200/tonne to plant gate.



Representation image



Representation image

Network Diagram:

See Hub-and-Spoke cluster map in Section 9 (distance table: Bookanakere 6 km, Chinakurali 9 km, Shravanabelagola 38 km). A GIS map with 10/20/35 km radius rings should be inserted before final issue. KML of all VCC locations to be attached as Appendix C.

FEEDSTOCK MATERIAL RECEIPT & PROCESSING

11. FEEDSTOCK MATERIAL RECEIPT & PROCESSING

A. FEEDSTOCK CHARACTERISTICS & MIXING STRATEGY

To maintain a consistent 1 TPH feed per reactor, the processing line must "homogenize" these three distinct materials:

Feedstock	Primary Challenge	Benefit to Biochar
Coconut Fronds	Extremely "woody" and tough; requires high-torque shredding.	Produces high-density, "hard" biochar with long soil life.
Sugarcane Trash	High volume, low density; prone to "nesting" in hoppers.	High carbon stability; excellent surface area.
Paddy Straw	High Silica content; can cause "slagging" in reactors.	Increases water-holding capacity of the biochar.

The "Mandya Mix" Strategy: For optimal reactor performance, a blend of **40% Cane Trash, 40% Paddy Straw, and 20% Coconut Fronds** is recommended to balance the throughput and biochar quality.

2. ELABORATED RECEIPT & PROCESSING WORKFLOW

A. Stage 1: Receipt & Material-Specific Handling

- **Coconut Fronds:** These arrive as long, bulky limbs. They must be diverted to a shredder before entering the main pulveriser line.
- **Cane Trash & Paddy Straw:** Arrive as bales. These are unbaled and fed into a pulveriser for size reduction for uniform flow.

B. Stage 2: Dual-Stage Shredding (The "Modular to 12 TPH" Path)

To achieve the 1 TPH flow rate, the material must be reduced to a uniform **10-20mm particle size**.

1. **Primary Shredding:** A slow-speed twin-shaft shredder breaks down the mixed coconut chips and straw/trash.
2. **Secondary Grinding:** A hammer mill refines the mix.
 - *Note for Scale-up:* when scaling to 6 plants (12 TPH), a centralized **Hammer Mill bank** with a pulse-jet dust extraction system is vital to prevent "dust explosions" from dry paddy straw.

C. Stage 3: Specialized Pre-Conditioning (Drying & Cleaning)

- **Vibratory De-Stoner:** This is crucial for KR Pet. Paddy straw and cane trash often carry soil and stones. This unit uses vibration and air to "lift" the biomass while letting heavy stones/sand drop, protecting the reactors from internal abrasion.
- **Rotary Drum Dryer:** Coconut fronds can have high internal moisture. Using the **waste heat** from the pyrolysis units, the "Mix" is dried to a stable **10-12% moisture**.

3. SCALE-UP PLAN: FEEDSTOCK RECEIPT & PROCESSING (YEAR 1 TO 3)

As the plant scale-up from **2 TPH to 12 TPH**, the processing section must transition from "batch-fed" to "continuous automated flow."

Year 1: 2 x 1 TPH (The Learning Phase)

- **Manual Sorting:** Laborers manually remove large stones or plastic from the conveyor.
- **Modular Feeding:** Each 1 TPH unit has its own dedicated small hopper and screw feeder.

Year 2: 6 x 1 TPH (The Hybrid Phase)

- **Automated In-feed:** Introduction of a **Walking Floor Bunker**. Trucks tip directly into a floor that "walks" the biomass into the shredder, eliminating the need for JCBs/loaders for every ton.
- **Centralized Drying:** A single large rotary dryer services all 3 units.

Year 3: 12 x 1 TPH (The Industrial Standard)

- **SCADA Integration:** The entire processing line (from de-stoning to reactor feed) is monitored from a central control room.
- **Pneumatic Distribution:** Processed "Mix" is blown through pipes or high-speed drag chains to the 12 individual reactors, ensuring no "starvation" of the feed.

4. ENVIRONMENTAL & SOCIAL IMPACTS IN PROCESSING

- **Environmental:** The **Silica** in paddy straw can become airborne. A high-efficiency **Cyclone Separator** must be installed to capture this dust, which can actually be sold as a "silica-rich" additive or mixed back into the biochar.
- **Social:** Processing Coconut Fronds is loud. The KR Pet plant must include **Acoustic Enclosures** for the chippers to stay within the 75dB limit for industrial zones, protecting the hearing of local workers.

- **Ecological:** The "De-stoning" process creates a byproduct of soil and organic fines. This "waste" should be returned to local KR Pet farms as a soil conditioner, ensuring a **Zero-Waste** facility.

PROCESSED FEEDSTOCK STORAGE

12. PROCESSED FEEDSTOCK STORAGE

Phase 1 Storage Sizing (2 TPH, 90-day buffer): Daily intake 48 T/day × 90 days = 4,320 T. At 150 kg/m³ stacked to 6 m: ~800 m² footprint. With fire clearances: allow 1.0–1.2 acres for Phase 1. The Phase 3 (12 TPH) end-state figures below are for long-term planning only.

A. THE "BUFFER" LOGIC

To sustain a **12 TPH** (6-unit) operation during the 90-day monsoon (June–August), the storage facility must act as a strategic reserve.

Volume & Area Requirements (90-Day Buffer)

- **Daily Demand (12 TPH):** 288 Tons/Day.
- **90-Day Demand:** ~26,000 Tons.
- **Bulk Density of Mix:** ~150 kg/m³ (for pulverized mix of cane trash, straw, and coconut).
- **Required Storage Volume:** ~173,000 m³.
- **Land Area Required:** * With a safe stacking height of **6 meters** (to prevent internal heat buildup), you require approximately **30,000 m² (7.5 Acres)** of dedicated storage space on the leased land.

B. FAVORABLE STORAGE CONDITIONS

Pulverized biomass in Mandya's humidity requires more than just a roof; it requires active climate management.

- **Moisture Control:** Maintain moisture <**15%**. If moisture exceeds 20%, microbial activity starts, leading to "Dry Matter Loss" (energy value drops) and high risk of **spontaneous ignition**.
- **Ventilation:** Use **Ridge Vents** and industrial louvers. Continuous airflow is necessary to dissipate the heat generated by the natural respiration of the organic matter.
- **Floor Construction:** A **RCC (Reinforced Cement Concrete)** floor with a vapor barrier (Polyethylene sheet) is mandatory. This prevents groundwater from wicking into the bottom 1 foot of the pulverized material.
- **Fire Safety: * Compartmentalization:** Divide the 7.5 acres into 10–12 separate "Bays" with 3-meter high firewalls.
 - **Sensors:** Install thermal probes in each stack. If the internal temperature hits **70°C**, the stack must be moved and cooled immediately.

C. BIOMASS FLOW & INTERNAL LOGISTICS

To feed 12 reactors continuously, the flow must move from "Static Storage" to "Dynamic Feed" using a **FIFO (First-In, First-Out)** principle.

Sequence of Flow:

1. **Primary Storage (The Buffer):** Pulverized material is moved via front-end loaders into the 90-day storage bays.
2. **Internal Transport:** electrically operated 1 ton tipper trolleys will transports the processed biomass from the leased storage land to the main plant.
3. **Day Bins:** Each 2 x 1 TPH plant has a 4-hour surge bin to ensure the reactor never runs dry during loader shifts.

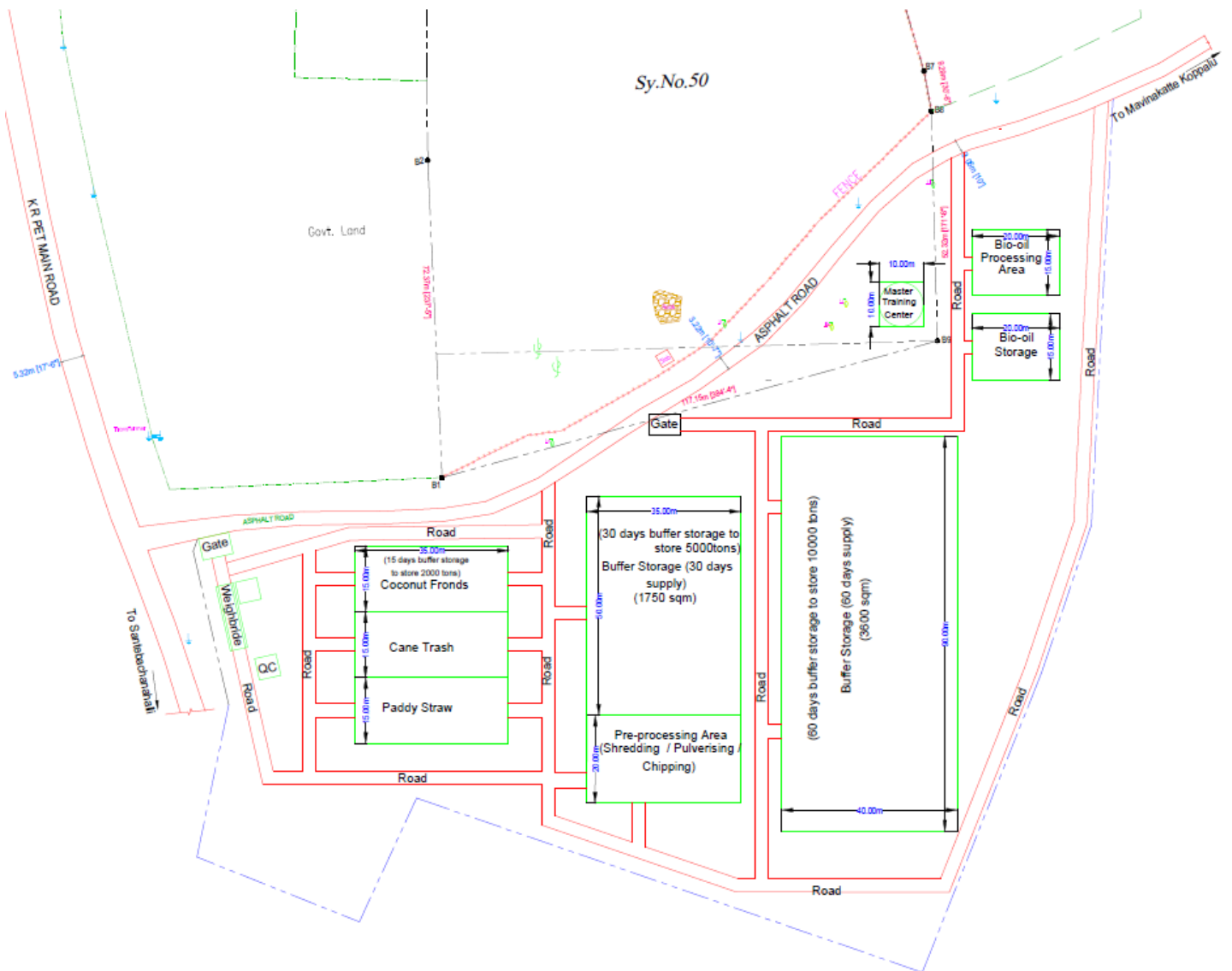
D. MONSOON CONTINGENCY PLAN (THE 90-DAY STRATEGY)

Risk Factor	Mitigation Strategy
No Field Collection	Use the "Active Bay" (Bay 1) while "Reserve Bays" (Bays 2–10) remain sealed under tarpaulin.
High Humidity	Recirculate warm air from the Pyrolysis Waste Heat through the storage floor ducts to keep the piles dry.
Logistics Interruption	Maintain a 48-hour "Ready-to-Feed" Silo directly at the plant site, separate from the main storage land.

E. INFRASTRUCTURE SUMMARY TABLE

Component	Year 1 (2 TPH)	Year 3 (12 TPH)
Storage Shed	5,000 sq. ft.	320,000 sq. ft. (7.5 Acres)
Handling	Manual + Small Loader	3 x 5-ton Payloaders + Automatic Reclaimers
Transport	Tipper trolleys	Automated Enclosed Conveyor Belt
Safety	Fire Extinguishers	Integrated Sprinklers + Thermal Imaging Cameras

F. Site layout map for the leased land

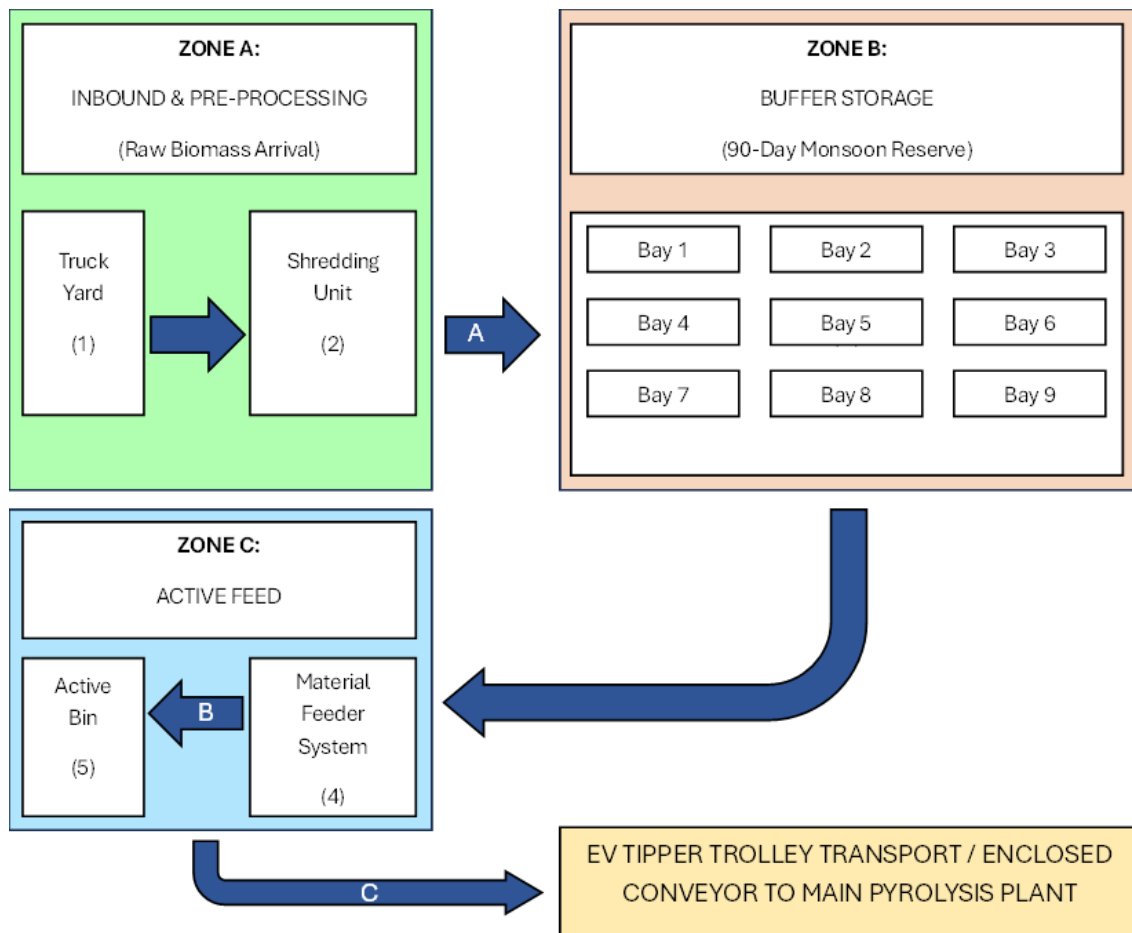


Note: not to scale

G. Site Layout Conceptual Map

The 5-acre leased land will be zoned to separate high-activity processing from long-term "quiescent" storage to minimize fire risk and dust interference.

LEASED LAND BOUNDARY - ~30,000 SQM



Key Flow Legend:

(1) Truck Yard: Unloading area for powdered coconut fronds & baled cane trash / paddy straw.

(2) Shredding Unit: Centralized pulverization point.

(A) Primary Conveyor: Transports fresh biomass to storage bays.

(3) Storage Bays: Reinforced 6m stacks with firewalls.

(B) Material Feeder: Retrieval of material for daily use.

(4) Reclaimer: Automated feed mechanism to ensure FIFO.

(5) Active Bin: Surge capacity (4-8 hours) for the plant.

(C) Plant Link: Final transit to the 12 TPH reactor line.

PYROLYSER OUTPUT - SUPPLY CHAIN MANAGEMENT

13. PYROLYSER OUTPUT - SUPPLY CHAIN MANAGEMENT

Biochar — Cooling, Quality Check, Bagging and Dispatch

Hot biochar exits the continuous pyrolysis reactor at 200–350°C via a water-jacketed screw conveyor (discharge screw with cooling jacket, 4 m length, supplied by Sanjivani Agro), which reduces temperature to below 60°C before the biochar enters the bagging zone.

Quality check: a representative 200 g sample is drawn from the discharge conveyor every 10 tonnes of production using an automated sampling arm. The sample is tested for moisture content (target $\leq 10\%$ at bag-in) and submitted for weekly full proximate analysis (carbon content, ash, moisture, volatile matter) by an ISO 17025-accredited off-site laboratory. Any batch with moisture $>15\%$ or visible uncharred material is diverted to the rework stream (recirculated through the dryer) and excluded from the carbon credit calculation for that batch. Approved biochar is packed in 50 kg PP woven bags, heat-sealed, and printed with a unique QR code encoding: Batch ID, production date, line number, weight, lab reference, and destination field parcel. Each bag is weighed on a digital in-line scale (accuracy ± 0.1 kg) and palletised (40 bags/pallet, 2 tonnes/pallet). Pallets are moved to the finished goods godown by a battery-operated pallet mover.

Finished goods buffer: minimum 30 days of production (~400 tonnes for Phase 1) maintained at all times.

Dispatch: barter biochar is delivered to village clusters in tractor-trailer runs (8–10 tonnes per trip); commercial surplus is dispatched on order with a Dispatch Note and Quality Certificate. All dispatch events are scanned in the Digital MRV system (Section 18) to complete the bag-to-field traceability chain.

DIGITAL M.R.V. & DASHBOARD

14. DIGITAL M.R.V. & DASHBOARD

Digital MRV (DMRV) tool support standardized, geotagged data collection across the biochar production value chain.

- a) Feedstock data: Source, type, quantity, moisture level, preparation process
- b) Production data: Production ID, pyrolysis temperature, GPS location, biochar characteristics
- c) Packaging data: QR codes linking each unit to production batch
- d) Application data: GPS location of field application, volume applied, mixed matrix

The core objectives of the proposed solution are:

- Implement full visibility from **biomass collection → biochar production → bagging → field application → SOC improvement → carbon credits**
- Build a scientifically compliant **carbon credit computation engine** (SOC-based, biochar-based, avoided emissions, etc.)
- Provide farmer- and parcel-level traceability to strengthen registry validation
- Ensure tamper-proof, verifiable data trails for carbon audits
- Efficiently manage carbon credit inventory, allocation, and trading
- Enable real-time dashboards and insights for both field operations and management

The solution includes 3 core layers:

A. Operational & Traceability Layer

1) Feedstock Collection & Transport

- Log biomass source details (generator, type, location)
- Capture trip details: loading points, weights, distances
- Estimate transport-based emissions
- Provide source-to-plant traceability

2) Biochar Production, QC & Batch Management

- Create production batches with process parameters
- Record QC attributes (moisture, temperature, residence time)
- Convert output into standardized **50 kg** QR-tagged bags
- Each bag uniquely mapped to:
 - Source feedstock
 - Batch
 - QC data
 - Destination

3) Bag Inventory & Dispatch

- Real-time bag-wise inventory
- Dispatch workflows to farmers / hubs / internal projects
- QR scanning at dispatch & receiving points
- Tamper-proof traceability from origin to application

B. Field Application & Agronomic Layer

1) Farmer & Land Parcel Management

- Maintain farmer profiles

- Record land parcels with GPS coordinates
- Track cropping pattern, soil type, acreage
- Support multiple parcels per farmer

2) Field Application Tracking

- Issue biochar bags to farmers
- Log bag-to-land parcel mapping
- Capture application events:
 - Photos
 - GPS
 - Date/time
 - Quantity per acre
- Mobile/PWA interface for ease of field operations

3) Soil Sampling & SOC Tracking

- Baseline soil tests for each parcel
- Periodic SOC re-tests after application
- Upload lab results & reports
- Compute SOC improvements automatically
- Maintain audit-ready SOC history

C. Carbon Credit & Registry Layer

1) Carbon Credit Computation Engine

The system will generate carbon credits based on:

- SOC delta (pre vs post application)
- Biochar carbon retention
- Avoided emissions (if applicable)
- Adjustment for transport emissions

The methodology and formulae will be configurable based on registry guidelines.

2) Carbon Credit Ledger & Allocation

- Track credits generated per farmer, project, region, season
- Allocate shares to:
 - Farmers
 - Shree Krishna CDR
 - Internal projects or partners
- Maintain a transparent ledger for all carbon credit movements

3) Registry Submission Workflow

- Create registry-ready data packets
- Track validation, verification & certification statuses
- Store and link all supporting documentation required by the registry
- API integration (if registry provides access)

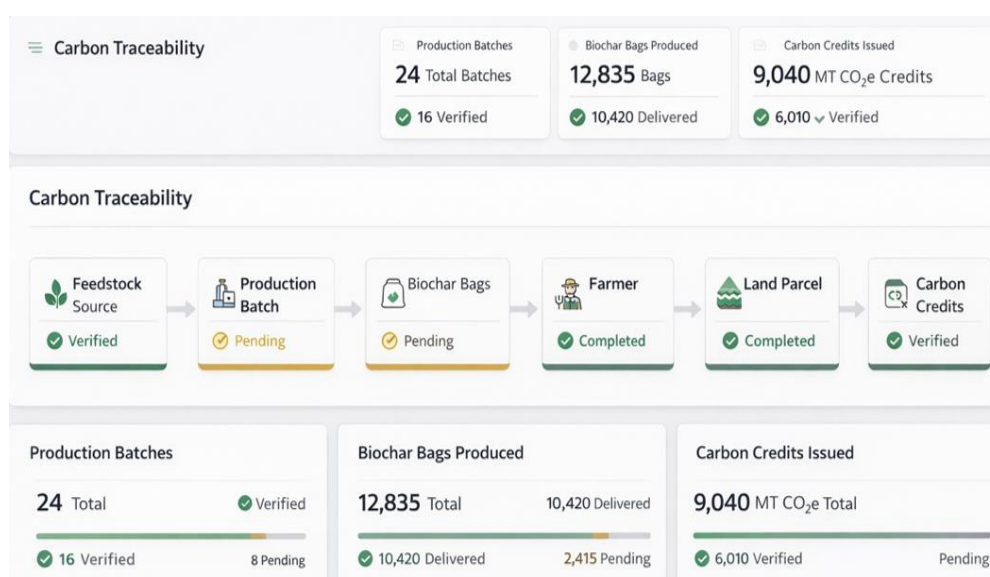
4) Carbon Marketplace / Buyer Management

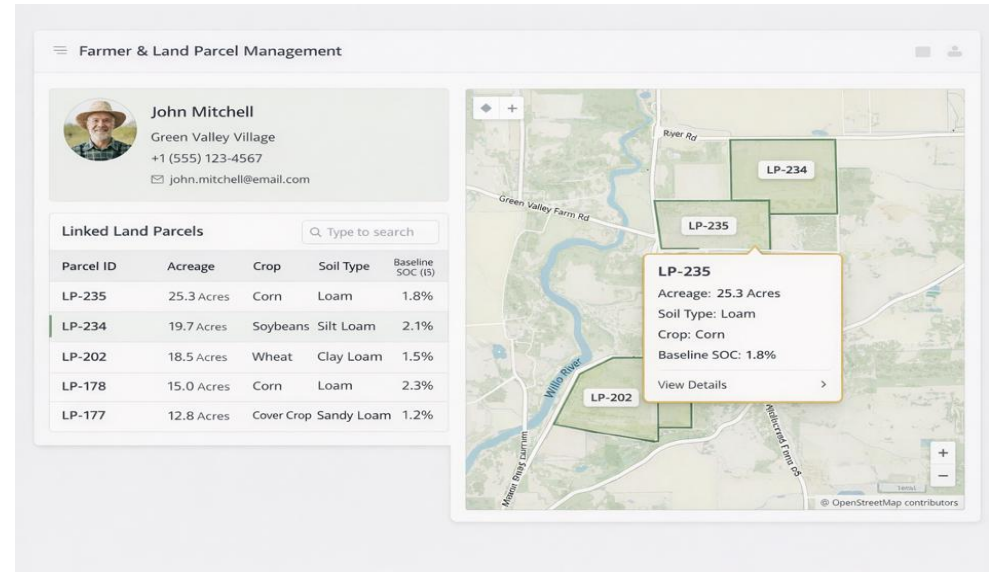
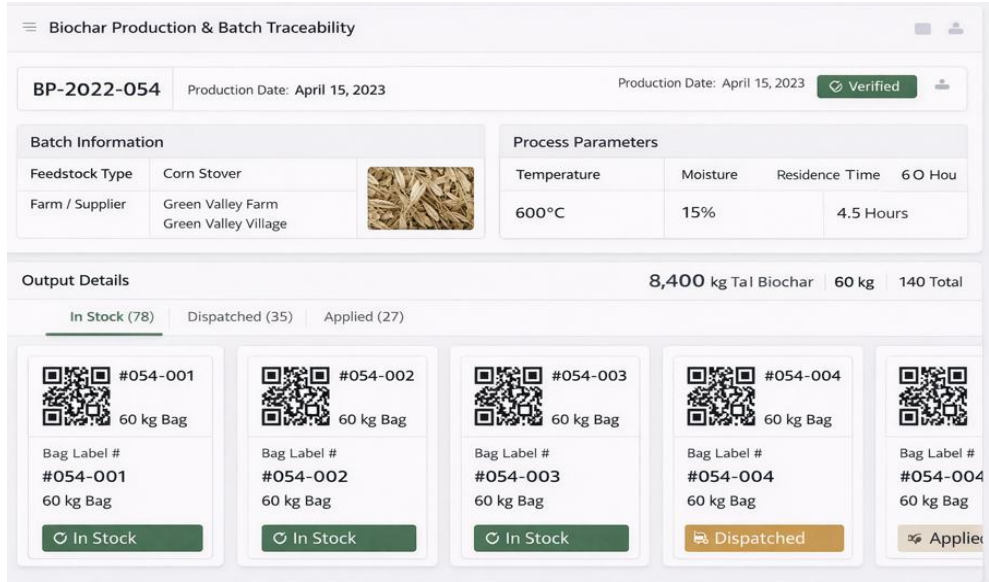
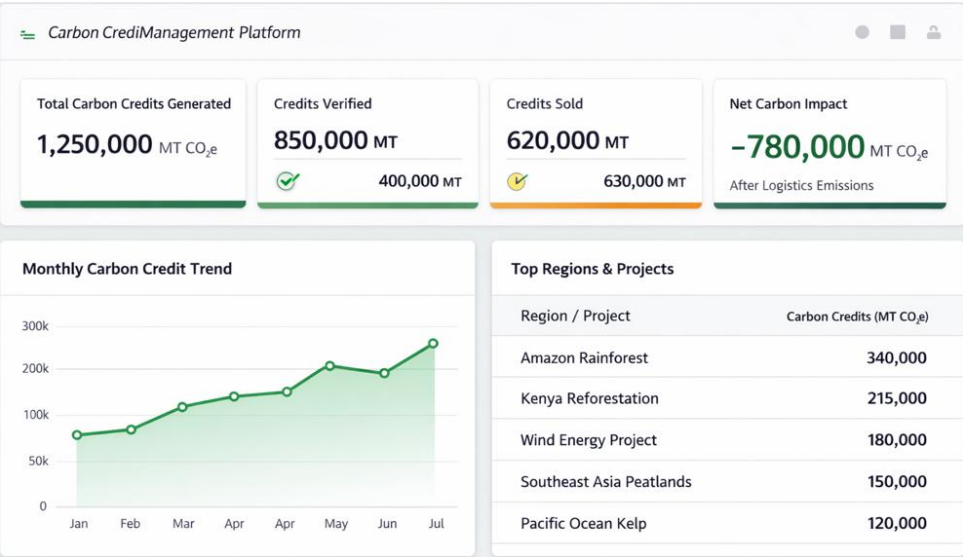
- Maintain inventory of verified credits
- Allocate credits to buyers
- Track transactions and ledger entries
- Generate invoices & buyer documentation

Blockchain-enabled transparency:

- Immutable data from biomass source to biochar sequestration to carbon credits issuance
- Digital MRV and Digitech Technician to collect & aggregate data, validate and time-stamp each step
- REAL-TIME MONITORING PLATFORM – live dashboard for investors & buyers
- Performance metrics: tCO₂e removed, hectares restored, biochar pH & C-content
- Remote auditing, automated reporting for carbon registries

Sample MRV Dashboard screens





BIO-OIL

15. BIO-OIL

15.1 Bio-Oil Yield and Characteristics

Bio-oil (pyrolysis oil) is the liquid condensate produced when the vapours and gases released during pyrolysis are cooled through the water-cooled condensers. For the mixed agricultural residue feedstock used in this project (coconut fronds, cane trash, paddy straw), the expected bio-oil yield is approximately 10–14% by weight of dry feedstock, with a design yield of 12%. At Phase 1 capacity of 14,400 TPA dry feedstock intake, annual bio-oil production is estimated at approximately 1,440–1,728 TPA. Bio-oil characteristics for agricultural residue pyrolysis: calorific value 15–19 MJ/kg (approximately 40% of diesel); pH 2–4 (acidic, corrosive — requires chemical-resistant storage); water content 15–30%; viscosity 50–1,000 cP at 20°C; density approximately 1.1–1.2 kg/litre. Bio-oil produced at 500–550°C separates into two fractions in the condenser train: an aqueous phase (upper layer) which is wood vinegar, and an organic phase (lower layer) which is the bio-oil proper. Both fractions are collected separately in dedicated collection vessels designed by Sanjivani Agro as part of the pyrolysis system package.

15.2 Storage Requirements

Due to its acidity and risk of phase separation, bio-oil must be stored in SS-304 or SS-316 stainless steel tanks with HDPE-lined interiors, in an RCC-bunded tank farm area (described in Section 18). Phase 1 storage capacity: 3 × 10,000-litre SS tanks, providing approximately 15 days of production buffer (~33 tonnes at 12% yield from 48 T/day feedstock intake). Tanks must be kept sealed to prevent water absorption (bio-oil is hygroscopic and phase-separates if water content exceeds 35%). Ambient temperature storage acceptable in Karnataka's climate (15–40°C); no heating required. Bio-oil must not be stored in carbon steel, aluminium, or copper vessels due to acidity-driven corrosion.

15.3 Offtake Options and Revenue Benchmark

Bio-oil from agricultural residues is currently marketed in three primary applications:

- (1) Industrial fuel supplement — blended with heavy fuel oil (HFO) for kilns and boilers at local brick kilns, ceramic units, and food-processing industries. Market price: approximately Rs 8,000–14,000 per tonne (benchmark; update with confirmed offtake agreement).
- (2) Asphalt bio-binder — the bio-oil residue (after the lighter fraction evaporates) forms a viscous bio-binder used in road construction as a partial replacement for bitumen. This is referenced in the main DPR's road construction plans for the plant perimeter roads.
- (3) Chemical feedstock — certain fractions (phenol, acetic acid, furfural) are of interest to specialty chemical buyers; however, this requires further processing and is a long-term Phase 3 opportunity. Indicative Phase 1 annual bio-oil revenue: 1,500 TPA × Rs 10,000/tonne = Rs 1.5 Crore/year (before transport and dewatering costs). Wood Vinegar: the aqueous condensate fraction (estimated 5–8% of feedstock, or 720–1,152 TPA at Phase 1) contains acetic acid, methanol, and phenols. Diluted 1:300 with water, it is applied as a plant bio-stimulant. A portion is distributed free to barter farmers as part of the soil improvement package. Surplus is sold to nurseries and organic farmers at approximately Rs 40–80 per litre (Rs 40,000–80,000 per tonne). Indicative Phase 1 annual wood vinegar revenue: 800 TPA × Rs 50,000/tonne (liquid, 1,000

litres/tonne) = Rs 4 Crore/year at full commercial sale (conservative estimate after barter allocation).

Source: ICAR-CIAE; Pyreg GmbH product data; Biochar Journal (2022) — pyrolysis liquid products from agricultural residues.

POWER SOURCING & ELECTRICAL BALANCE OF PLANT (BOP)

16. POWER SOURCING & ELECTRICAL BALANCE OF PLANT (BOP)

Power Sourcing Strategy

To ensure the plant operates at 24/7 capacity, a hybrid power sourcing model is implemented to mitigate the risks of grid instability in the rural K R Pet/Hassan region.

- **Grid Connectivity:** A dedicated **11kV or 33kV Express Feeder** line from the nearest KPTCL (Karnataka Power Transmission Corporation Limited) substation. This minimizes voltage fluctuations that can damage sensitive VFDs (Variable Frequency Drives) in the pulverizing unit.
- **Captive Power (Bio-Energy):** Utilization of the **Syngas** produced during the pyrolysis process. A portion of this gas is redirected to a **Syngas Generator (Genset)** to produce "Green Power," reducing the net grid dependency by **30-40%**.
- **Solar Integration:** Installation of ground mounted Solar PV panels on the **~40,000 sq. ft** biochar plant area. This power is used primarily for the **Logistics Control Tower**, lighting, and office utilities, ensuring that data tracking continues even during a total grid blackout.
- **Battery Energy Storage System (BESS):** A containerized BESS will be installed to stabilize power to bridge the gap during grid fluctuations or outages, ensuring the C&I system and critical cooling pumps never lose power. Store excess solar energy during the day to offset peak-hour grid costs.

Electrical Balance of Plant (BOP) Components

The BOP encompasses all electrical infrastructure required to support the core pyrolyzer and the auxiliary supply chain machinery.

- **Main Receiving Substation (MRS):** Includes the Outdoor Switchyard, 2.0 MVA Transformers (Step-down from 11kV to 415V), and SF6 Circuit Breakers.
- **Motor Control Centers (MCC):** Dedicated MCC panels for the **Biomass Pre-processing Wing**. These panels manage the high-inertia start-up of the Hammer Mills and Primary Shredders.
- **Variable Frequency Drives (VFDs):** Critical for Supply Chain Management. VFDs are installed on all **Screw Conveyors** and **Rotary Feeders** to precisely control the "Dosing Rate" of feedstock into the kilns, allowing the plant to adjust to variations in biomass density.

Automation & SCADA Integration

The Electrical BOP is the hardware foundation for the **Logistics Control Tower (LCT)**.

- **PLCs (Programmable Logic Controllers):** Installed at every node of the supply chain—from the Intake Weighbridge to the Final Biochar Bagging unit.
- **Energy Monitoring System (EMS):** Provides real-time data on "Energy per Ton of Biochar." If the energy consumption of a Hammer Mill spikes, the LCT alerts the O&M team to check for blade wear or high-moisture feedstock.

- **Uninterruptible Power Supply (UPS):** A centralized Industrial UPS system supports the PLC/SCADA and LCT servers, ensuring that no data regarding feedstock "Proof of Origin" is lost during power transitions.

Electrical Load Distribution (Estimated for 12 TPH)

Load Category	Estimated Power (kW)	Criticality for Supply Chain
Pre-processing (Hammer Mills/Shredders)	600 - 800 kW	High: Primary bottleneck if power fails.
Material Handling (Conveyors/Elevators)	150 - 200 kW	High: Essential for "Feedstock-to-Kiln" flow.
Pyrolysis Core (Fans/Blowers/Screws)	300 - 400 kW	Vital: Core production unit.
Auxiliary (LCT/Lighting/Pumps)	50 - 80 kW	Medium: Essential for safety and tracking.

Safety & Earthing Systems

Given the high dust environment (Coconut/Paddy dust), electrical safety is paramount:

- **Hazardous Area Classification:** All electrical motors and switchgear in the pulverizing and bagging zones must be **Explosion-proof (Flameproof - Ex-d)** to prevent dust-ignition incidents.
- **Lightning Protection:** Specialized lightning arrestors for the storage warehouse and the 12 TPH plant stacks to protect against the high-intensity electrical storms.
- **Redundant Earthing:** Copper-plate earthing for all PLC/SCADA panels to prevent signal interference from high-power motors.

AUTOMATION & INSTRUMENTATION

17. AUTOMATION & INSTRUMENTATION

17.1 Field-Level Automation (Sourcing & Collection)

In restricted village spaces, large balers cannot enter. Automation here focuses on miniaturization and mobility.

- **Mini-Baler Automation:** Use of compact, automated pick-up balers for paddy straw and cane trash that can be towed by small (35-45 HP) tractors common in Mandya.
- **IoT-Enabled Handheld Moisture Meters:** Bluetooth-integrated probes that automatically upload moisture readings (MC%) to the central cloud before a load is even cleared for pickup.
- **Geo-Fencing & Mobile App Logging:** A custom sourcing app for field officers that uses GPS to “tag” the biomass origin. This is the first step in the **MRV (Monitoring, Reporting, and Verification)** chain.

17.2 Village Collection Center (VCC) Instrumentation

The VCC acts as the “Pre-Processing Hub” where bulk is converted into manageable units.

- **Automated Weight Bridges (Small Scale):** Pitless, low-profile weighbridges (5–10 tonne capacity) that automatically log weight into the ERP via RFID tags assigned to each tractor.
- **Industrial Shredder Automation:** Automated feed-rate controllers for coconut frond shredders. These sensors adjust the belt speed based on motor torque to prevent jamming and ensure uniform particle size for the pyrolyzer.
- **Solar-Powered IoT Weather Stations:** To monitor ambient humidity and temperature, predicting the “natural drying time” required for sugarcane trash piles.

17.3 Transportation & Logistics Optimization

Restricted road space requires a “Milk-Run” logistics model rather than bulk hauling.

- **Dynamic Route Optimization Software:** An AI-driven dispatch system that calculates the best routes through narrow village roads, avoiding school hours or local market days, and ensuring tractors are not idling.
- **RFID/QR Code Tracking:** Every bale or crate of shredded fronds is assigned a QR code. When scanned during loading/unloading, it updates the “In-Transit” inventory in real-time.
- **Live Fleet Telematics:** GPS tracking on all collection tractors to monitor “Diesel Liters per Tonne-KM,” helping to reduce the Scope 3 emissions of the supply chain.

17.4 Biochar Plant Gate & Feedstock Management (Proposed)

- **Automated Sampling Robots:** A mechanical “sampler” that takes core samples from incoming loads to check for ash content and internal moisture using **Near-Infrared (NIR) Spectroscopy**.

- **Ultrasonic Level Sensors:** Installed in the plant's storage silos to automatically trigger a "Sourcing Request" from the VCCs when levels drop below 20%.
- **Automated Conveyor Sorting:** Magnetic separators and pneumatic cleaners to automatically remove stones, metal wires, or plastic from the biomass before it enters the pyrolyzer.

17.5 MRV (Monitoring, Reporting, & Verification) Instrumentation

To sell high-value Carbon Removal Credits (CORCs), the MRV must be "Tamper-Proof."

- **Distributed Ledger Technology (Blockchain):** All data from the field (moisture, weight, GPS, time) is hashed into a blockchain. This creates an immutable "Carbon Audit Trail."
- **Automated Mass-Balance Accounting:** Software that cross-references the **Weight In (Biomass)** vs. the **Weight Out (Biochar)**.
 - *If 5 tonnes of biomass produce 1.5 tonnes of biochar, the system automatically calculates the CO₂ sequestration value based on fixed carbon sensors in the reactor.*
- **Digital Flow Meters:** To measure the syngas and bio-oil byproducts, ensuring the "Net-Negative" energy balance is maintained and verified.

Summary Table: Automation Impact

SCM Stage	Tool/Instrument	Benefit for Restricted Space	MRV Impact
Field	Mini-Baler / GPS App	Manoeuvrability in narrow bunds	Proof of Origin
Transport	Route AI / RFID	Prevents traffic bottlenecks	Logistics Emission Data
VCC	NIR Moisture Probes	Accurate "Dry Weight" calculation	Prevents "Water Weight" fraud
Plant Gate	Automated Sampling	Faster turnaround of tractors	Quality Assurance

CIVIL & STRUCTURAL WORKS; BUILDINGS

18. CIVIL & STRUCTURAL WORKS; BUILDINGS

Site Preparation & Infrastructure

The leased 5-acre site must be graded and compacted to handle heavy machinery and EV tipper traffic, especially during the monsoon.

- **Land Grading & Hardstanding:** 70% of the area should be developed with **WMM (Wet Mix Macadam)** or **PCC (Plain Cement Concrete)** flooring to prevent biomass contamination with soil and to support the weight of industrial chippers and grinders.
- **Perimeter Fencing:** 2.4m high chain-link fencing with 3 strands of barbed wire and a 10m wide main gate for heavy truck/tractor entry.
- **Stormwater Drainage:** Intensive peripheral drainage is mandatory to prevent water logging around the biomass piles, which can lead to spontaneous combustion.

Civil Works for Processing & Weighment

- **Weighbridge Foundation:** A pitless, RCC foundation for a 20 MT electronic weighbridge (approx. 3m x 6m). A dedicated 100 sq. ft. **Weighbridge Cabin** for the operator and SCM software hub.
- **Processing Shed (Industrial Shed):** A 10,000 sq. ft. PEB (Pre-Engineered Building) structure with a minimum height of 8m to accommodate:
 - **Machine Pedestals:** Reinforced RCC pads for the **Chippers, Grinders, and Pulverizers** to dampen vibrations.
 - **EV Charging Bay:** Dedicated civil platforms with fire-rated walls for charging **EV Tipper Trolleys**.
 - **Dust Extraction Trenches:** Sub-surface trenches to manage fine dust generated during grinding.

Storage Facilities (Processed & Buffer)

The most critical structural challenge is storing the 3-month buffer stock (~4,500 tonnes).

- **Daily Consumption Shed (Processed Feedstock):** A 5,000 sq. ft. enclosed shed near the plant inlet. It must have **anti-static flooring** and be completely moisture-proof.
- **3-Month Buffer Storage (Raw/Baled):**
 - **Open Plinths:** Raised RCC plinths (0.5m above ground level) to store baled paddy straw and cane trash.
 - **Vertical Silos (Optional for Pulverized Material):** To save space, pulverized biomass can be stored in 20-ton vertical steel silos with automated discharge.
- **Bio-Oil Storage & Decantation Area:** * **Containment Bunds:** An RCC "Dyke Wall" area (approx. 1,000 sq. ft.) to house SS-304/316 bio-oil tanks. The bund must be able to hold 110% of the largest tank's volume in case of a leak.
 - **Decantation Pit:** A specialized pit with an oil-water separator to prevent bio-oil runoff into the ground.

Water & Utilities

- **Borewell & Raw Water Storage:** An RCC or Sintex-banked platform for a **1,00,000-liter Raw Water Tank**. This serves the pyrolysis cooling system and fire hydrants.
- **Fire Hydrant System:** A dedicated ring-main pipeline around the biomass storage area with high-pressure monitors. Biomass yards are high-fire-risk zones; civil works must include an **Underground Fire Sump**.

Summary of Building Requirements

Building Unit	Area (Approx.)	Construction Type	Purpose
Admin & MRV Office	1,500 sq. ft.	Masonry / RCC	Lab, Control room, and Documentation.
Main Processing Shed	10,000 sq. ft.	PEB / Steel Column	Housing Chippers/Grinders/Pulverizers.
Buffer Storage Yard	2.5 Acres	Hardstanding Plinths	3-month raw biomass stock.
Daily Feed Shed	5,000 sq. ft.	Enclosed PEB	Processed dry feedstock for immediate use.
Bio-Oil Depot	2,000 sq. ft.	RCC Bunded Area	Tank farm and decantation.
Labor Rest Area	800 sq. ft.	Masonry	Shift staff amenities and toilets.

Structural Design Considerations for MRV

To meet **MRV (Monitoring, Reporting, and Verification)** standards, the civil works must facilitate "Batch Segregation." The storage plinths should be partitioned into "Lots" with unique ID markers. This allows auditors to verify that the biomass processed in "Line 1" on a specific day matches the sourcing data from a specific village cluster.

HEALTH, SAFETY & ENVIRONMENT MANAGEMENT

19. HEALTH, SAFETY & ENVIRONMENT MANAGEMENT

A. SAFETY POLICY

1.1 Purpose of Safety Organisation

- This organization is responsible for maintaining safety during the complete contract period of a project. The basic approach to maintain safety is as follows:
- All personnel employed on a project are responsible for their own safety and the safety of personnel next to them.
- Minimization of near-miss accidents can result in minimization of possible actual accidents.

1.2 Responsibilities of Safety Organisation

1.2.1 Site Manager- Safety Manager

Site Manager is responsible for the safety of all activities related to the project. His responsibility is not limited to the safety of the construction work in the field but also for all transportation of personnel, project materials, health and hygienic conditions of work at site and accommodation.

Site Manager is responsible for Safety activities.

1.2.2 Safety Engineer

Safety Engineer is responsible for organizing the safety sites related to the project. He makes timely visits to the construction activities and reports to site manager. Safety engineer receives all data of near-miss accidents from the construction superintendent and analyses those in order to advise the construction superintendent to take necessary action for reducing and minimizing those near-miss accidents.

1.2.3 Construction Superintendent- Safety Superintendent

Construction superintendent is responsible for the safety of the actual activities of all construction for the safety of the actual activities of all construction work, over and above the related activities of work, including all transportation of personnel, project materials, offices and accommodation etc. Construction superintendent gives all necessary instructions to have safe working conditions for personnel and monitors the same during construction period of the project. The construction superintendent implements all requirements of safety and reports and result achieved. All safety documents. Safety manual, safety instructions, safety reports, accidents reports, etc are established by construction superintendent, verified by safety engineer and approved by site manager.

1.2.4 Safety Management Team

Site manager organizes the safety management team which consists of the site manager, safety engineer, and construction superintendent. The safety

management conducts safety meetings twice a month to discuss safety requirements, unsafe works found during the past period and to promote safe working and safety requirements for all construction activities until completion of project.

1.2.5 Safety Committee

Site Manager organizes the safety committee which consists of the safety management team plus the quality control engineer, field planning/progress monitoring engineer, material controller and the medical officer.

Safety committee holds meeting once in a month to discuss safety requirements, unsafe work found on site, suggestion for improving and maintain safe working conditions on construction site. The safety committee makes spot check safety patrol to ensure safety.

1.2.6 Safety Representatives

All persons engaged in any activity of the project are responsible for their own safety and safety of others working in the neighbouring areas. Safety representatives carry out their work within the safety regulations ensuring that they provide no hazards to safety of others and observe the instructions they receive from the safety supervisor.

1.3 Safety Instructions

All personnel engaged on the project are responsible for performing and monitoring all construction and related activities in accordance with the following safety regulations and manuals.

- Fire and safety regulations
- Minimum health standards in contractor's camp for construction workers
- Requirement of wearing safety helmets on construction site
- Safety policy
- Safety procedures on construction site
- General safety requirements
- Work adjacent to existing plant facilities
- General safety rules

2.0 SAFETY PROCEDURE

The main purpose of the safety procedure is to carry out work safety in a manner so as to eliminate hazards and risk of death or injury to our personnel, and to third parties on or off the work site and of damage to the works and other facilities and equipment. Also to eliminate the risk to personnel of long term injury due to health hazards. Safety communication chart is attached hereto, are followed for the implementation of Safety procedures.

2.1 Project Manager

Project Manager is responsible for safety of all activities related to the area, not limited to only construction work in the area but also for all transportation

of personnel, project materials, health and hygienic conditions of work at site and accommodation.

2.2 Safety Engineer

Safety Engineer assists in organizing the safety activities. He makes timely visits to the construction sites to check all working conditions of construction activities and any deviation from safety rules and regulation is reported to site - in – charge for correction. Safety Engineer receives all data of accidents from Project Manager and he prepares the detailed safety statistics for submission to Plant Owner. He analyses the safety statistics in order to advice the concerned site-in-charge to take necessary actions for minimizing the frequency of the accidents.

2.3 Site in-Charge

Site-in-charge is responsible for the safety of the actual activities of all construction work and other related activities of work, including all transportation of personnel and materials, and running offices and accommodation camp etc. site- in- charge makes all necessary instructions to have safe working conditions for the work and personnel and monitors these during the entire period of the projects. Site-in- charge receives instructions and advice from the Project Manager and the Safety Engineer for safety and takes necessary measures in accordance with the instructions and reports on the results achieved.

2.4 Safety Supervisors

Every engineer, supervisor, foreman, and other staff including office workers, assumes the duties of safety supervisor for the activities which they are handling or supervising, safety supervisors take all necessary actions for the safety of activities under their supervisions. Safety supervisors in addition to assuming responsibility for their own activities, liaise with other supervisors carrying out work in the same area, to ensure that the activities of one team do not endanger the safety of another team.

2.5 Safety Representatives

All persons engaged in any activity of the project are responsible for their safety of their own actions and for the safety of others working in the neighbouring area. Safety representatives ensure the safety of others and ensure that the instructions received from the safety supervisors are observed.

- 2.6 All work adjacent to the existing plant facilities are carried out strictly under the safety procedure formulated by Plant owner. All necessary precautions are observed to protect the facility well as to carry our job in a safe way.

- 2.7 Camp and working areas are protected by providing fire fighting arrangements at all possible fire hazard areas. Most of our employees are trained to use fire extinguishers. Periodical training is organized to train our new employees. Regular maintenance of fire extinguishers is carried out.
- 2.8 Periodical fire and safety training Programmes are organized for educating all our employees. Construction hazards and their prevention are highlighted during the training. Some of our employees are deputed to attend the safety training Programmes organized by client or other organizations.
- 2.9 Hard hats, safety shoes, hand gloves, etc., are issued to our employees free of cost for their personal protection. Specific type of safety equipments are provided free of cost to employees as and when required by the employees.
- 2.10 All potential lost time, fatal or vehicle accidents or near miss accidents are immediately reported to Plant owner's safety engineer. Detailed investigation of all accidents is carried out. The causes of accidents and their preventive measures are circulated among all our sites to prevent the recurrences of similar type of accident.

Accident is reported to client through the standard proforma.

- 2.11 Monthly safety statistics are prepared and submitted to the Plant owner through the proforma.
Any specific safety report is submitted as and when required by Plant owner.

Annual summary of safety statistics is submitted to Plant owner.

- 2.12 Any anomalies with regard to safety matters is resolved at site. Specific job safety procedure, if required, is prepared.
- 2.13 Maintenance workshop is established at all our camp sites to carry out periodical maintenance of our vehicles and equipment to keep them in good condition. Periodical vehicle checks are carried out to eliminate traffic accidents. Selected supervisors are deputed to attend the defensive driving course organized by safety organizations or Plant owner to issue our internal company's driving license to successful drivers. A drivers incentive scheme has been introduced.
- 2.14 A good house keeping committee consisting of the Manager (Project), Safety Engineer, Admin. Officer and medical representative is formed. Housekeeping committee inspects our camp and site facility periodically and inspection report is prepared. Inspection report is

sent to the concern site-in-charge for immediately implementation of any suggestions made.

- 2.15 In case of any emergency, immediate actions jointly agreed with Plant owner are taken by the site management has sole authority to prepare emergency action plans.
- 2.16 Any suggestion which helps us to keep our site safe, clean, hygienic and secured is implemented immediately to maintain our good safety standard.
- 2.17 Fire A\and safety regulations are followed strictly at all our working sites to achieve our goal of accident free completion of project.

Works Adjacent to Existing Plant Facilities

The lighting of fires and the introduction of matches, cigarette lighters and other sources of ignition close to the existing oil and gas facilities is strictly forbidden.

All work, especially hot work, excavation and entry to vessels is controlled by a permit –to-work system.

Industrial Compressed Gas Cylinders

Gas cylinders are transferred, stored and utilized as per the standard safety practices mention in fire and safety rules and regulations.

Excavation

To prevent damage to underground services and to protect the excavating crew, all excavation in or close to company facilities is carried our under the excavation permit procedures formulated by Plant owner.

Work requiring dogging of trenches is programmed so that trenches are not left open for longer than essential.

Personnel working in trenches or pits, excavating soil, clay, sand and similar materials are protected against earth slides, falling objects excessive water build-up trenches. All pits, trenches and excavation are roped off. Warming lights are provided if any excavation is kept open overnight.

Fire Protection and Site Cleanliness

All the works, adjacent facilities, stores, camps and other facilities are protected from damage of fire by providing, inspecting and maintaining suitable types of fire extinguishers.

3.0 GENERAL SAFETY RULES

In addition to the safety regulations mentioned in the previous chapter, the following general safety rules are followed at the construction sites:

- a. Safety helmet and safety shoes are worn.
- b. Supervisor is present at worksite especially where heavy and hazardous jobs are being carried out.
- c. Electrical equipment is considered live unless it is ensured that it is isolated and made dead.
- d. No switch protected with danger board/tag is operated.
- e. Tampering with any types of valves, equipment or moving machinery at site is prohibited.
- f. Water is prevented from falling on electrical equipment
- g. All electrical equipment is properly earthed.
- h. All aisles, passageways, and stairways are kept clean and unobstructed.
- i. It is forbidden to remove guards from moving machinery or to clean/check moving parts of machinery while in motion.
- j. Starting of any machines or any power tools is done only after ensuring that nobody is in danger.
- k. Gas cylinders are kept away from the direct rays of the sun, hot places, and live electric lines. Valves on such cylinder are never lubricated.
- l. It is forbidden to stand under a suspended load

4.0 GENERAL SAFETY REQUIREMENTS

4.1 Introduction and Responsibilities

All work In-charges / Supervisors carry out the work safely to eliminate hazards and to minimize the risk of death or injury to his own personnel, to Client personnel and to third parties on or off the site, and of damage to the works and other facilities and equipments. Supervisors also eliminate the risk to personnel of long term injury due to health hazards.

Supervisor remains fully and exclusively responsible for ensuring that the works are carried out in a safe manner.

4.2 Role of Safety Engineer

To advise on all safety matters, to run safety training, to set up and maintain safety promotion and incentive schemes and to carry out safety inspections. To control and provide liaison on all safety matters with client.

4.3 Safety Training, Inspection and Schemes

Periodical fire and safety training programmes are arranged. Some employees are nominated to attend any specific safety training programmes organized by Plant Owner.

Frequent safety inspections are carried out to find unsafe acts, conditions and concerned supervisors are informed to avoid such unsafe acts and conditions.

Safety Incentive Scheme for the workers has been formulated. Safety Incentives scheme for Drivers has already been formulated and being followed.

4.4 Reporting and Data Gathering

Supervisor immediately reports verbally all potential lost time, fatal or vehicle accidents or near-miss accidents to Safety Engineer for submission to Owner. A detailed investigation report of all lost time accidents is prepared and submitted to the plant owner.

All accidents are reported to Plant owner on a standard Accident Report Form.

Monthly Safety Statistics are submitted to Plant owner on a Standard statistics form.

4.5 Vehicles and Driving

Special attention is given to all matters of road safety with the aim of reducing road accidents and fatalities.

DRIVERS Each driver has a valid Driving License for the type of vehicles he is driving, Maximum speed limit of 80 KMPH is observed and lower speed limits as stipulated in the areas like Camp and Plant adhered to.

All drivers and front seat passengers wear seatbelts.

Vehicles having adequate capacity for the required duty and which are designed for the service assigned are utilized. No vehicle is driven in an unsafe condition or by an unlicensed driver. All Vehicles are properly identified with the Company's name and serial number. Wheels, tools

(including those required to change a wheel), adequate spare fuel, spare water etc., are provided for each journey.

Unless operationally essential, the driver keeps to the good graded or hard topped road system.

Plant site speed limit: 20 km/h. Loaded delivery vehicles require a banksman when reversing. No vehicle to be left running within 10 m of covered biomass storage. [Replaced from inapplicable EPC template]

- Good house keeping is maintained. All unnecessary inflammable and other surplus material are removed from the site or facility, Sufficient waste bins are provided on work site.
- Safe storage for all inflammable substances is provided in safe location.
- Vehicle parking: biomass delivery vehicles must use the designated weighbridge holding bay only.
- All the scrap and rubbish is disposed of at a location and in a manner as approved by plant owner.

4.6 Provision and Operation of Plant, Tools and Equipments

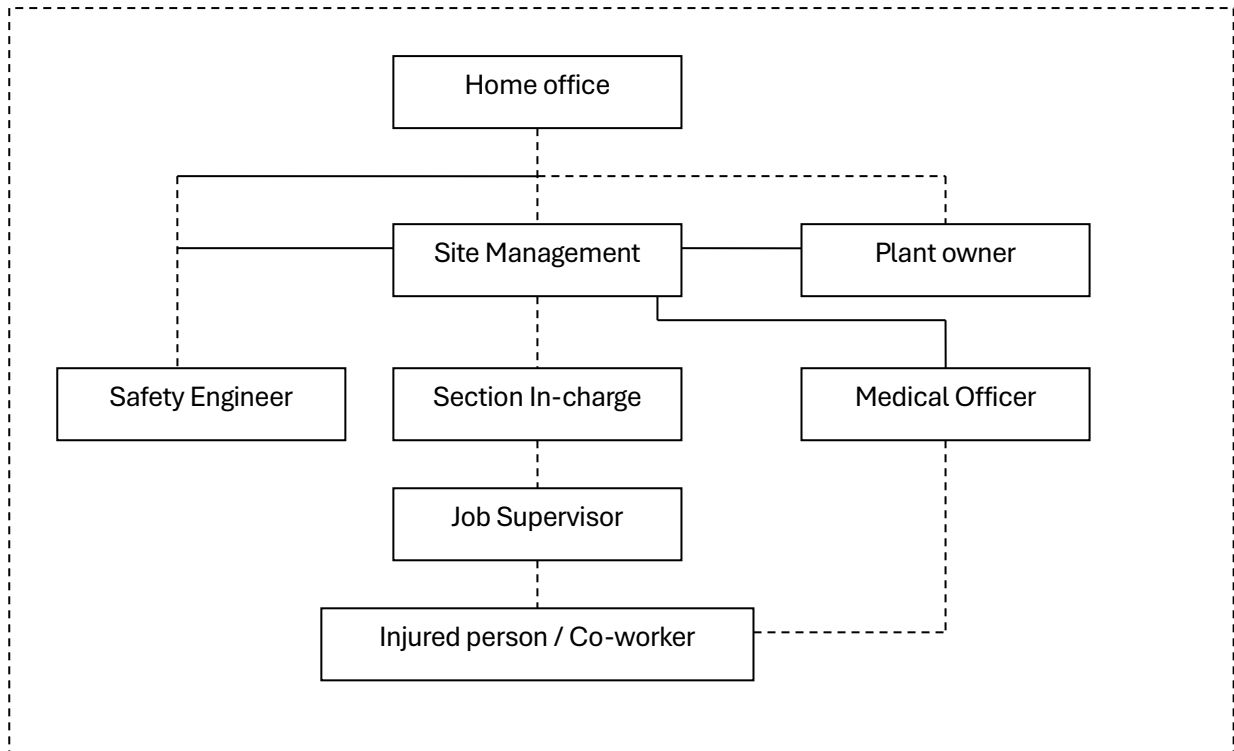
All plant, tools and equipments are maintained in operable condition. Users of the plant, tools and equipment are trained, experienced, and where necessary, licenced or certified. Topmost safety standard is maintained as far as lifting equipments and electrical equipments are concerned.

4.7 Protective Equipments

Protective clothing and equipment are provided to employees and it is ensured that they use these. Adequate eye and body protection is provided to men engaged in welding, burning and grinding operations. Respiration protection equipments are provided to employees working in air contaminated area like shotblasting area, spray painting area etc.

All Safety gear is used as per the demands of the job.

Safety Communication Chart



ACCIDENT REPORT CHART

_____ Reporting Line

----- Communication Line

Environmental Management

5.0 ENVIRONMENT POLICY

5.1 *Policy objectives*

Plant supports clients establish an Environment Policy for their organization which would encompass

- Climate Change Impact Assessment
- Environment Impact Assessment

5.2 *Customer Support Organisation*

Plant provides the above customer support services through its sister company Grameena Abhivrudhi Mandali (GAM) which is a Section 25 Company having mission of sustainable Development of rural communities (refer website: www.gam-rd.co.in).

GAM, as per mandate of Government of India, has developed/is developing various Distributed Generation projects based on locally available agricultural residues/surplus produce, animal waste, food processing/industry waste, etc.

A key element of GAM initiatives are

- Proper understanding of GHG (Green House Gases) mitigation through using Bio Resources / Waste as feedstock for Energy Systems.
- Proper understanding of CHP (Combined Heat & Power) solutions that could be adopted as part of Bio Energy Schemes
- Significant empirical knowledge of producing Organic Fertiliser (using waste from Biomass/Biogas Power Plants) as well as its application for different crops/different soil conditions.
- Significant empirical knowledge in implementing sustainable development projects in rural areas and evaluating socio-economic impact.

5.3 Climate Change Impact Assessment

- Displacement of electricity from the grid by generating electricity from Biomass / biogas and exported to the grid.
- Reduction in open-field burning: by diverting agricultural residues to pyrolysis instead of field burning, the project reduces PM2.5, CO, and CO2 emissions in K R Pet taluka during the October–November cane trash and May–June paddy straw burning seasons, consistent with Karnataka KSPCB air quality targets.
- The biomass waste used by the project activity would have been left to decay anaerobically in a solid waste disposal site without methane recovery in the baseline case. Hence, methane emissions are avoided.

5.4 Environment Impact Assessment:

5.4.1 Environmental Assessment Procedure

- Collection of relevant data from various sources such as site visits by a surveyor to the proposed project area, interviews with various stakeholders identified by the project and collection of primary and secondary data from government records on agro-forestry, agriculture and agriculture related industries.
- A desktop review and compilation of this data.
- Environmental aspects (project activities that had potential to impact the environment) of the project will be identified. e.g. Emissions of flue gas

from stack, fossil fuel emissions from transportation of biomass, generation of effluent from the water treatment plants etc.

- The environmental aspects will be prioritized based on their potential to have adverse impacts on the project environment and its stakeholders.
- Environmental management plans will be formulated, to control / manage / mitigate these impacts to the best extent possible. The primary aim of the environmental management plan will be to ensure that regulatory standards were met (also reflected in the Power Plant design) and that there exists appropriate monitoring procedures for collection of the relevant data.

5.4.2 Management of Potential Environmental Impacts

- The power plant and auxiliaries shall be ensured to meet the emission norms of the National Ambient Air Quality Standards (NAAQS).
- No radioactive emissions are expected from the proposed Biomass /Biogas plant.
- A green belt, of minimum 5 m width, shall be provided around the periphery of the power plant.

5.4.3 Identification of Socio-economic aspects

Current socio-economic profile of the project area

a. Main income generating activities	
b. Main income generating assets	
c. Average Income of a Farmer from an acre of land per annum	
d. Average level of education	
e. Main types of local transport	
f. Locally available natural resources	

Based on the current socio – economic profile a detailed assessment will be conducted to identify the various potential socio-economic aspects related to the project. These aspects are as listed below:

- Production of quality and reliable power.
- Power generation through a renewable energy technology.
- Activities related to the biomass supply chain i.e. transportation, supply and preparation of biomass.
- Employment requirements at the project site

Socio-economic benefits that will be evaluated

- Increase in productivity of agricultural land and underutilized lands.
- Employment generation through sustainable biomass supply chain

- Grama Panchayats role and involvement.
- SHGs role and involvement.
- Employment generation through organic fertilizer unit and cold storage.
- Employment generation in Biomass/ Biogas Power Plant.
- Employment generation through growth of micro-enterprises, facilitated by availability of assured quality/reliable electric power
- Dissemination of information, to local community, on the advantages of renewable energy-based technologies and their applications in creating local value-addition.
- In general, an improvement in the quality of life of the local people, living within the project boundaries.

***ORGANIZATION –
PLANT
CONSTRUCTION AND
O&M***

20. ORGANIZATION – PLANT CONSTRUCTION AND O&M

Construction Phase: Supply Chain Infrastructure

The construction of the plant is divided into two parallel streams: the **Pyrolysis Units** and the **Biomass Processing Infrastructure**.

A. Civil & Structural Engineering

- **Feedstock Storage Bunkers:** Construction of ~40,000 sq. ft. of reinforced concrete flooring with heavy-duty load-bearing capacity (to handle heavy duty material movement equipment's).
- **Preprocessing Sheds:** Elevated steel structures with noise-dampening cladding to house the Primary Shredders and Hammer Mills.
- **Fire Suppression Network:** Installation of an underground ring-main hydrant system with high-capacity pumps to cover the 90-day biomass buffer zone.

B. Machinery Installation & Commissioning

- **Intake Systems:** Installation of dual 10-ton pitless weighbridges and hydraulic truck tippers for rapid unloading of pulverized coconut fronds, cane trash & paddy straw.
- **Conveyor Integration:** Mechanical assembly of enclosed drag-chain and screw conveyors linking the storage bunkers to the day bins.
- **Utility Setup:** Construction of a dedicated transformer yard and backup power (DG sets) specifically rated to handle the high-torque surge of the pulverizing units.

Operations & Maintenance (O&M) Framework

Post-construction, the O&M team ensures the "12 TPH feed demand" of the plant is consistently met through a high-availability supply chain.

A. Integrated Operations Management

The O&M is governed by a **Feedstock-First Policy**. The plant's output is directly throttled by the quality of input material.

- **Feedstock Conditioning:** O&M staff are responsible for "Blended Feeding"—mixing high-lignin coconut fronds with high-silica paddy stalks to maintain the specific fixed-carbon profile required by customers.
- **Shift Synchronization:** Maintenance windows for the pyrolysers are staggered with the warehouse cleaning cycles to ensure 24/7 uptime.

B. Preventive Maintenance Schedule (Supply Chain Assets)

To prevent logistical bottlenecks, a rigorous PM schedule is implemented for the "First-Mile" equipment (this will strictly follow the manufacturer / supplier maintenance schedule):

- **Mobile Units:** Bi-weekly blade sharpening and hydraulic checks for mobile pulverisers in KR Pet and Nagamangala clusters.

- **Plant Machinery:** Daily inspection of conveyor belt tension and weekly lubrication of Hammer Mill bearings.
- **Magnetic Separators:** Shift-wise cleaning of the over-band magnets to ensure no metal ingress enters the pyrolyser screws.

Supply Chain Organizational Hierarchy

The O&M organization is led by the **Plant Head**, with a dedicated vertical for **Logistics and Feedstock Security**.

Key O&M Verticals:

1. **Sourcing & Procurement Wing:** Responsible for relationship management with farmers & stakeholders in K R Pet and Channarayapatna; manages the "Gate Price" based on market fluctuations.
2. **Fleet & Logistics Wing:** Manages the "Milk-Run" schedules, fuel monitoring via the LCT, and driver coordination for the 25km radius.
3. **Process & Quality Wing:** Operates the preprocessing mills and ensures the feedstock moisture and particle size are within the prescribed limits (<20% moisture, 5-8 mm size).

Health, Safety, and Environment (HSE) in O&M

Given the combustible nature of the feedstock and the dust generated during pulverizing, HSE is integrated into the O&M manual:

- **Dust Mitigation:** Use of misting systems and industrial vacuum units in the shredding zone to keep Ambient Air Quality (AAQ) within Karnataka State Pollution Control Board (KSPCB) norms.
- **Fire Drills:** Mandatory monthly fire-suppression drills for all warehouse staff, focusing on "Deep Pile" fire management.
- **Personal Protective Equipment (PPE):** Strict enforcement of N95 masks, anti-static footwear, and ear protection in the preprocessing bays.

Scaling Strategy (O&M Integration)

As the plant moves from **2 TPH to 12 TPH**, the O&M structure scales via **Modular Expansion**:

- **Phase 1 (Current):** Centralized O&M.
- **Phase 3 (Scale-up):** Decentralized O&M, where "Cluster Managers" are stationed at the Village Hubs to handle local equipment maintenance, reducing the need to transport broken balers back to the main plant.

IMPLEMENTATION SCHEDULE

21. IMPLEMENTATION SCHEDULE

Assuming the "**Start Date**" (**Day 0**) is when the feed to the biochar plant begins, the following 6-month (24-week) forward planning schedule is required to establish a smooth supply of the feedstock.

Phase 1: Identification & Mapping (Weeks 1–4)

- **Feedstock Mapping (Week 1-2):** Map "hotspots" of sugarcane trash and paddy straw within a 20–30 km radius of the KR Pet plant site to minimize transportation costs.
- **Chemical Analysis (Week 3):** Sample biomass from different clusters to test for moisture content and ash levels. High moisture in Mandya's sugarcane trash requires specific drying protocols.
- **Supplier Shortlisting (Week 4):** Identify large-scale farmers, sugar mills (for bagasse/trash), and local green entrepreneurs.

Phase 2: Stakeholder Engagement & Legal Framework (Weeks 5–10)

- **Stakeholder Awareness Drives (Week 5-7):** Conduct "Kiran Melas" or village-level meetings in KR Pet taluka to explain the benefits of selling residue vs. burning it.
- **Agreement Drafting (Week 8):** Finalize **Feedstock Purchase Agreements (FPA)**. These should include barter agreement, price-per-ton (indexed to moisture) and quality standards.
- **Contract Signing (Week 9-10):** Target 150% of the required tonnage (approx. 2,200 tons/month) in signed contracts to account for seasonal volatility and farmer "leakage."

Phase 3: Logistics & Infrastructure Setup (Weeks 11–16)

- **Logistics Tendering (Week 11-12):** Identify local transporters with tractor-trailers or 10-wheelers. Prioritize those with "baling" equipment (crucial for cane trash & paddy straw).
- **Logistics Agreements (Week 13-14):** Sign **Service Level Agreements (SLA)** with transport providers. Include clauses for "Turnaround Time" at the KR Pet plant.
- **Collection Centers (Week 15-16):** Set up 2–3 decentralized "Satellite Collection Points" in the taluka where farmers can drop off biomass if they don't have transport.

Phase 4: Buffer Stocking & Testing (Weeks 17–23)

- **Procurement Kick-off (Week 17):** Begin physical collection of biomass.
- **Buffer Building (Week 18-22):** Accumulate a **30-day buffer (approx. 1,500 tons)** at the plant site. This is critical for KR Pet to survive the monsoon or harvest gaps.
- **Pre-Processing Check (Week 23):** Run the shredding and drying units (the "front end" of the 1 TPH lines) to ensure the feedstock meets the size and moisture requirements of the pyrolysis reactor.

Phase 5: Operational Launch (Week 24 / DAY 0)

- **Commissioning:** Start the 2 x 1 TPH feed.
- **Supply Chain Stabilization:** Transition from "Buffer Stocking" to "Just-in-Time" delivery for 70% of the feed, keeping 30% from the buffer.

Timeline Summary Table

Milestone	Duration	Primary Objective
Mapping & Lab Tests	Month 1	Confirm biomass quality in KR Pet.
Stakeholder Agreements	Month 2	Secure the raw material at fixed costs.
Logistics Setup	Month 3	Procure Shredders, balers and transport contracts.
Buffer Construction	Month 4-5	Storage of 1,500 tons to prevent plant downtime.
Feed In (Go-Live)	Day 0	Continuous 2 TPH operation begins.

Key Social & Ecological Guardrails

- **Ecological:** Ensure the agreement mandates that 10–15% of residues remain on the farm for soil health (in situ decomposition).
- **Social:** Implement a digital payment system for KR Pet farmers to ensure transparency and "payment within 7 days" to build trust.

STORAGE REQUIREMENTS FOR 1,500-TON BUFFER (MONSOON READY)

In KR Pet, monsoon rainfall can exceed 200mm, "open storage" is a risk for 1 TPH reactors which require <20% moisture for efficiency.

Technical Infrastructure Needed:

- **Land Requirement:** Approximately **1.5 to 2 acres** specifically for the 1,500-ton bale stack (accounting for spacing and utilities).
- **Flooring:** **Stone-dust or brick-lined base** with a slight slope so as not to store biomass directly on red/clay soil during monsoon; to avoid moisture absorption from the ground.
- **Covering:** High-grade **UV-stabilized Silpaulin (250+ GSM)** or permanent steel sheds. Tarpaulins must be lashed securely to withstand the high wind speeds of the pre-monsoon.
- **Fire Safety:** Install a **Fire Hydrant system** or a dedicated water tanker (10,000L).

- Install **Lightning Arrestors** on the storage yard, as biomass stacks are high-risk targets during monsoon thunderstorms.

LOGISTICS & QUALITY CONTROL CHECKLIST

Activity	Standard Operating Procedure (SOP)
Shredding & Baling	Field-dry biomass to <20% moisture before baling.
Transport	Use a "First-In, First-Out" (FIFO) system at the plant gate.
Moisture Check	Every truck must be tested with a digital moisture probe.
Stack Monitoring	Weekly temperature checks of the buffer stacks to prevent spontaneous combustion (internal heat buildup).

PROJECT COST BREAKDOWN

22. PROJECT COST BREAKDOWN

Civil Works

Sl No	Item	Unit	Qty	Rate / unit (INR)	Amount (INR)
	Civil Works				
1	Layout preparation, levelling, filling and Primary roads formation	Sqm	16,000	75	12,00,000
2	WBM Roads formation 5m wide	Rm	950	1,000	9,50,000
3	Storm Drains and utility drains	Rm	950	700	6,65,000
4	PEB Process Buildings - 1No 40 M X 45M - 1600 Sqm Including foundations PCC	Sqm	1,600	10,000	1,60,00,000
5	Reservoir 25 KLD capacity Raw water and treated water	Lts	25,000	20	5,00,000
6	Fire sump 100 KLD	Lts	1,00,000	18	18,00,000
7	STP 10KLD	Lts	10,000	30	3,00,000
8	Administrative building	Sqm	200	5,000	10,00,000
9	Lab Building	Sqm	200	5,000	10,00,000
10	Main control Building	Sqm	200	5,000	10,00,000
11	Security Room	Sqm	20	3,000	60,000
12	Water and sewerage systems	LS	1	20,00,000	20,00,000
13	Raw material Storage room	Sqm	2,000	5,000	1,00,00,000
14	Fencing	Sqm	1,040	1,000	10,40,000
			Total Civil Works		3,75,15,000

Electrical & Mechanical Works

Sl No	Item	Unit	Qty	Rate (INR)	Amount (INR)
	Electrical Mechanical works				
1	Water pumps, Fire pumps	Lot	1	20,00,000	20,00,000
2	Fire Hydrants and Extinguishers	Lot	1	10,00,000	10,00,000
3	Transformers (330KVA)	No	1	8,00,000	8,00,000
4	Main Control Panel	No	1	20,00,000	20,00,000
5	Plant control Panel	No	1	3,00,000	3,00,000
5	Cameras	Lot	1	5,00,000	5,00,000
6	DGA 110KVA	No	1	10,00,000	10,00,000
7	Cabling	Lot	1	20,00,000	20,00,000
				Total	96,00,000

PROJECT FINANCIALS OVERVIEW

23. PROJECT FINANCIALS OVERVIEW REQUIRED

23.1 Supply Chain OPEX — Phase 1 (2 TPH, 14,400 TPA Gross Feedstock)

Feedstock Procurement (Barter + Cash Split Model): Under the confirmed barter model, farmers who supply 4 tonnes of dry biomass receive 1 tonne of biochar in return. This barter eliminates a direct cash outflow for the biochar portion of farmer compensation. However, the barter model is supplemented by a cash logistics margin paid to LGEs and transporters for collection, pre-processing, and first-mile delivery costs.

Indicative cash OPEX for feedstock and logistics: transport and logistics (tractor-trailers, LCVs, diesel): approximately Rs 800–1,200 per tonne delivered; LGE service fee (logistics margin above barter): approximately Rs 300–500 per tonne dry; at-source processing (chipper/baler hire, fuel): approximately Rs 200–400 per tonne.

Total estimated feedstock procurement and logistics OPEX: Rs 1,300–2,100 per tonne of dry biomass delivered to plant gate. For 10,010 TPA net dry feedstock: annual logistics OPEX approximately Rs 1.3–2.1 Crore per year.

Note: the biochar portion of the barter ($4,800 \text{ TPA} \div 4 = 1,200 \text{ TPA}$ biochar distributed as barter) represents an opportunity cost equivalent to $1,200 \text{ TPA} \times \text{market price of biochar}$ — this must be accounted for separately in the financial model as a barter contra-revenue line.

Plant OPEX (Supply Chain Infrastructure): operating labour for feedstock handling (see Sections 9 and 20 for shift structure: 9 persons per shift $\times$ 4 teams = 36 staff): approximately Rs 60–80 lakhs per year; preventive maintenance for at-source fleet and plant-gate processing equipment: approximately Rs 15–20 lakhs per year; MRV platform operational cost (digital tools, cloud hosting, IoT data): approximately Rs 10–15 lakhs per year; power for pre-processing (shredders, hammer mills, conveyors): approximately $150\text{--}200 \text{ kW} \times 6,000 \text{ hrs/year} \times \text{Rs } 7/\text{kWh} = \text{Rs } 63\text{--}84 \text{ lakhs per year}$ (partially offset by captive solar and syngas genset). Total estimated annual supply-chain OPEX (Phase 1): Rs 2.5–3.5 Crore per year.

23.2 Supply Chain Revenue — Phase 1

The supply chain generates three revenue streams at Phase 1:

(1) Carbon Removal Credits (primary revenue): Phase 1 Module 1 net CDR $\sim 8,971 \text{ tCO}_2/\text{year}$; target carbon credit price to be confirmed via B2B offtake agreement (indicative range USD 100–150/tCO₂ per Puro.earth market benchmarks, Q1 2026). Indicative carbon revenue: $8,971 \text{ tCO}_2 \times \text{USD } 120 \times \text{Rs } 83/\text{USD} = \text{approximately Rs } 8.9 \text{ Crore per year at full operation}$.

(2) Bio-Oil: approximately $1,500 \text{ TPA} \times \text{Rs } 10,000/\text{tonne} = \text{Rs } 1.5 \text{ Crore/year}$ (indicative).

(3) Wood Vinegar (commercial surplus after barter allocation): approximately $500 \text{ TPA} \times \text{Rs } 50,000/\text{tonne} = \text{Rs } 2.5 \text{ Crore/year}$ (indicative). (4) Commercial Biochar Sales: 4,800 TPA produced; 1,200 TPA barter; 3,600 TPA surplus. At Rs 10,000–12,000/tonne: $\sim \text{Rs } 3.6 \text{ Crore/year}$. Total indicative annual revenue at Phase 1 steady state: approximately Rs 16.5–18 Crore per year (Carbon Credits Rs 8.9 Cr + Bio-Oil Rs 1.5 Cr + Wood Vinegar Rs 2.5 Cr + Commercial Biochar Rs 3.6 Cr).

Total indicative annual OPEX: Rs 2.5–3.5 Crore (supply chain only; add main plant OPEX for full picture). Supply chain contribution margin: approximately Rs 9–11 Crore per year at Phase 1.

Note: All numbers above are indicative benchmarks pending contract confirmation.

23.3 Supply Chain CapEx (Phase 1)

The supply chain CapEx for Phase 1 (2 TPH, 14,400 TPA gross feedstock) covers at-source processing fleet, logistics vehicles, Village Collection Centre infrastructure, and MRV digital tools, in addition to the plant-gate civil and electrical works already costed in Section 26.

Indicative supply chain CapEx: at-source mobile chippers (3 × Rs 3–4 lakhs) = Rs 9–12 lakhs; round balers with rakes (2 × Rs 9–12 lakhs) = Rs 18–24 lakhs; tractor-trailers (4 × Rs 14–18 lakhs) = Rs 56–72 lakhs; LCV for long-haul and sampling (1 × Rs 12–15 lakhs) = Rs 12–15 lakhs; VCC shed structures (2 sites × Rs 15–20 lakhs) = Rs 30–40 lakhs; IoT moisture meters and GPS trackers (Rs 3–5 lakhs); MRV software platform setup (Rs 10–15 lakhs).

Total indicative supply chain CapEx: Rs 1.38–1.83 Crore. Combined with the Section 26 plant civil and electrical CapEx (Rs 4.71 Crore), total Phase 1 project CapEx is estimated at approximately Rs 6–7 Crore (USD 720,000–840,000). [Update all line items with confirmed vendor quotes.] Data source for benchmarks: Karnataka State Agricultural Marketing Board (KSAMB) tractor and implement hire rate schedule 2025; MNRE/NCEF biomass processing equipment market survey 2024.

RISK ASSESSMENT & MITIGATION PLAN

24. RISK ASSESSMENT & MITIGATION PLAN

Risk management is a core requirement for the long-term credibility, bankability, and carbon integrity of the Biochar Project. Unlike conventional industrial projects, biochar-based carbon removal initiatives face **multi-dimensional risks** and managed by classifying the risk.

A. Risk Classification

Risks are categorized into the following groups:

- a) Feedstock and supply chain risks
- b) Technical and operational risks
- c) Environmental, health, and safety (EHS) risks
- d) Regulatory and legal risks
- e) Carbon methodology and MRV risks
- f) Financial and market risks
- g) Social, ESG, and reputational risks
- h) Force majeure and external risks

a) **Feedstock & Supply Chain Risks** cover

- **Feedstock Availability Shortfall:** due to seasonal variability, competing uses, or climatic factors that could reduce biomass availability.

The Mitigation measures considered are multi-feedstock strategy (coconut fronds, cane trash, paddy trash, bagasse, bamboo, mixed residues), village-wise diversified sourcing across villages spread across KR Pet. Stock a buffer inventory of ~30 days

- **Feedstock Price Volatility:** increased demand for residues could drive up procurement costs.

The Mitigation measures considered are long-term sourcing arrangements, localized sourcing to reduce transport cost & continuous productivity improvement in logistics

b) **Technical and operational risks**

- **Technology Underperformance:** Lower-than-expected output or biochar yield could impact production and credit volumes.

The Mitigation measures considered are proven pyrolysis technology, OEM performance guarantees, conservative design capacity assumptions and gradual ramp-up and stabilization period

- **Equipment Failure or Downtime:** Breakdowns could disrupt operations and carbon delivery.

The Mitigation measures considered are Preventive and predictive maintenance systems, critical spare parts inventory, redundancy in key systems and trained maintenance teams

- **Operational Skill Gaps:** Inadequate operator skills could affect safety and consistency.

The Mitigation Measures considered are structured training by **Skills Council for Green Jobs (SCGJ)**, certification and re-certification systems and SOP-driven operations

c) **Environmental, Health, and Safety (EHS) risks**

- **Fire and Thermal Hazards:** Biomass and high-temperature operations pose fire risks.

The Mitigation Measures considered are Fire detection and protection systems, controlled biomass storage, emergency shutdown systems, Regular drills and training

- **Environmental Non-Compliance:** Failure to meet emission or waste norms could lead to penalties or shutdown.

The Mitigation Measures considered are multi-stage pollution control systems, regular environmental monitoring & compliance audits

- **Occupational Health Risks:** Exposure to dust, heat, or noise could affect workers.

The Mitigation Measures considered are Engineering controls, PPE and health surveillance & Job rotation

d) **Regulatory and legal risks**

- **Delay or Change in Regulatory Approvals:** Delays in statutory approvals could impact project timelines.

The Mitigation Measures considered are Early and proactive regulatory engagement & periodic review of compliance requirements

- **Policy or Legal Changes:** Changes in environmental or industrial regulations.

The Mitigation Measures considered are review of Compliance requirements & Continuous policy monitoring

e) **MRV risks**

- **MRV Data Gaps or Errors:** Incomplete or inaccurate data could affect credit issuance.

The Mitigation Measures considered are Digital MRV systems, Trained MRV personnel & Internal audits and cross-checks

- **Permanence and Reversal Risk:** Risk of carbon reversal over time.

The Mitigation Measures considered are Production of stable biochar, Application in appropriate end-uses & Conservative permanence assumptions

f) **Financial and market risks**

- **Cost Overruns:** Unexpected increases in CAPEX or OPEX.

The Mitigation Measures considered are Contingency provisions, Cost control systems & Performance-based contracts

g) **Social, ESG, and reputational risks**

- **Community Opposition:** Potential resistance from local communities.

The Mitigation Measures considered are No land displacement, Local employment generation & Stakeholder engagement mechanisms

- **ESG Non-Alignment:** Failure to meet buyer ESG expectations.

The Mitigation Measures considered are Strong HSE and HR systems, Transparent reporting & Social and environmental safeguards

h) **Force majeure and external risks**

- **Extreme Weather Events:** Floods, droughts, or storms affecting operations.

The Mitigation Measures considered are Robust site design and drainage, Insurance coverage & Emergency preparedness plans

- **Supply Chain Disruptions:** Fuel shortages or transport disruptions.

The Mitigation Measures considered are Local sourcing, Multiple transport options & Inventory buffers

B. Risk monitoring, Review & Governance

a) **Governance Structure**



C. Risk Monitoring & Review Mechanism

- Periodic risk reviews by management (Monthly → Internal, Quarterly → Management review, Annual → Third-party audits)
- Integration with internal audit systems
- Updating mitigation measures as needed

Risk management is treated as a **continuous process**, not a one-time exercise.

***ENVIRONMENTAL,
ECOLOGICAL &
SOCIAL IMPACT***

25. ENVIRONMENTAL, ECOLOGICAL & SOCIAL IMPACT

The supply chain related to biomass sourcing & biochar supply has an impact on **Environmental** (physical/chemical shifts), **Ecological** (living systems/biodiversity), and Social (human/economic) dimensions.

In the context of the upcoming biochar plant in KR Pet Taluka, Mandya, the impacts are uniquely tied to the region's status as a "Sugar Bowl." The heavy reliance on sugarcane and paddy cultivation in this belt transforms the typical biochar discussion into a localized strategy for waste management and soil restoration.

Environmental impacts

Relating to the physical and chemical transformation of the Mandya landscape.

Upstream (Sourcing)

- **Mitigation of Residue Burning:** Farmers in Mandya often burn sugarcane trash (dry leaves) to clear fields for the next crop. A plant in KR Pet provides an alternative to this, significantly reducing local air pollution and CO₂ emissions in the taluka.
- **Water Sourcing for Pyrolysis:** While biochar production is a dry process, cooling systems for large plants require water. Given Mandya's reliance on the KRS Dam/Cauvery water, the plant's water footprint must be managed to avoid competing with irrigation needs.

Downstream (Supply & Application)

- **Alkalinity Management:** Soils in parts of Mandya can become alkaline due to intensive irrigation. Since most biochar is naturally alkaline, the plant must tailor the "recipe" (pyrolysis temperature) to ensure the biochar does not further spike soil pH.
- **Drought Resilience:** KR Pet often faces water stress during lean seasons. Biochar application in Mandya's red and black soils can increase water holding capacity (WHC), acting as a "sub-surface reservoir" for sugarcane crops.

Ecological impacts

Relating to the biological health and biodiversity of the KR Pet ecosystem.

Upstream (Sourcing)

- **Soil Micro-fauna Preservation:** By moving away from "field burning" to "factory sourcing," the beneficial soil bacteria and earthworms in KR Pet's fields are preserved rather than incinerated during the clearing process.

- **Biomass Balance:** Sourcing must ensure that not *all* trash is removed. Leaving a small percentage of organic matter is vital for the local ecosystem's natural decomposition cycle.

Downstream (Supply & Application)

- **Microbial "Hotspots":** The porous nature of biochar serves as a habitat for mycorrhizal fungi, which are essential for nutrient uptake in Mandya's paddy fields.
- **Reduction in Chemical Runoff:** Intensive farming in Mandya leads to high fertilizer runoff into local canals and the Hemavathi/Cauvery River system. Biochar application helps "lock" these nutrients in the soil, preventing eutrophication (algal blooms) in downstream water bodies.

Social impacts

Relating to the farmers, economy, and health of the people in KR Pet.

Upstream (Sourcing)

- **New Revenue Streams:** Traditionally, sugarcane trash is a liability for KR Pet farmers. The plant creates a "waste-to-wealth" model where farmers can sell this residue, providing a financial cushion against fluctuating sugar prices.
- **Logistics Employment:** The collection and transport of biomass from various villages in the taluka to the central KR Pet plant will create local green jobs in trucking and loading.

Downstream (Supply & Application)

- **Reduced Fertilizer Costs:** Mandya farmers are heavily dependent on urea and DAP. By improving nutrient efficiency, biochar can reduce the quantity of expensive chemical fertilizers needed per acre, increasing net profit for the farmer.
- **Respiratory Health:** Moving from open-field burning to controlled industrial pyrolysis will lead to a cleaner "burning season" in KR Pet, reducing the incidence of asthma and respiratory issues among the rural population.
- **Carbon Credit Dividends:** If the plant is linked to international carbon markets, there is potential for KR Pet farmers to eventually earn carbon credits for the stable carbon they "sink" into their land.